Nametable Mapping

File: nametable_mapping.kicad_sch

CPU Address Bus

File: cpuAddressBus.kicad_sch

Register Access

File: register_access.kicad_sch

PRG Banking

File: prg_banking.kicad_sch

PPU Address Bus

File: ppu_address_bus.kicad_sch

Reset Detection

File: reset_detection.kicad_sch

CPU Data Bus

File: cpu_data_bus.kicad_sch

PCM DAC

File: pcm_dac.kicad_sch

Scanline Detection

File: scanline_detection.kicad_sch

V-Split

File: v_split.kicad_sch

PPU Fetch Counting

File: ppu_fetch_counting.kicad_sch

CHR A0,A1,A2

File: chr_a0_a1_a2.kicad_sch

ExRAM Mode

File: exram_mode.kicad_sch

Fill Mode

File: fill_mode.kicad_sch

Pulse Length Lookup Table

File: pulse_length_lookup_table.kicad_sch

Pulse Timer

File: pulse_timer.kicad_sch

Pulse Duty Envelope Volume

File: pulse_duty_envelope_volume.kicad_sch

Pulse DAC

File: Pulse DAC.kicad_sch

PPU Data Bus

File: ppu_data_bus.kicad_sch

CHR Bank Registers

File: chr_bank_registers.kicad_sch

CHR Bank Selection Logic

File: chr_bank_selection_logic.kicad_sch

CHR Address Bus 10 to 14

File: chr_address_bus_10_to_14.kicad_sch

CHR Address Bus 15 to 19

File: chr_address_bus_15_to_19.kicad_sch

Multiplier

File: multiplier.kicad_sch

MMC5A Timer

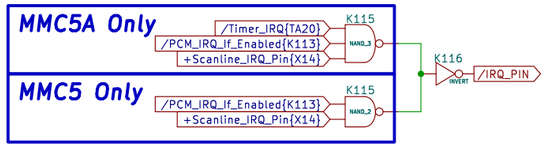
File: mmc5a_timer.kicad_sch

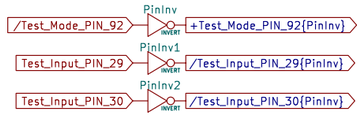
MMC5A CL3 SL3 GPIO

File: mmc5a_cl3_sl3_gpio.kicad_sch

Artifacts

File: artifacts.kicad_sch



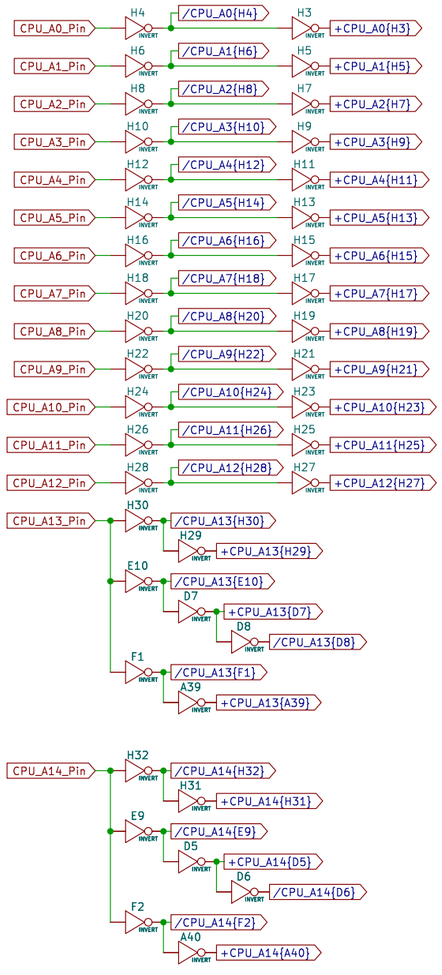


Sheet: /
File: MMC5.kicad_sch

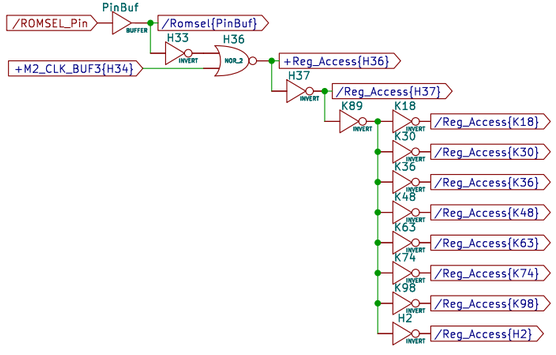
Title:

Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 1/28

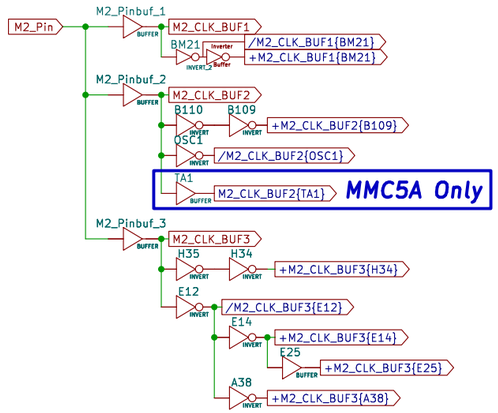
CPU Address Bus Buffering



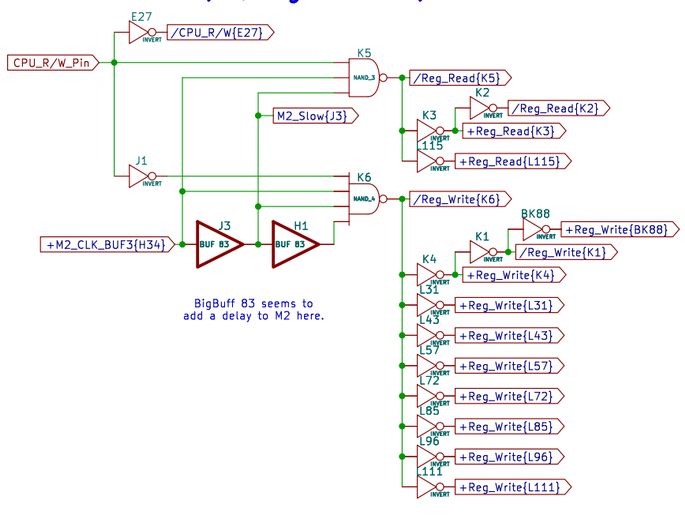
/ROMSEL, Register Access



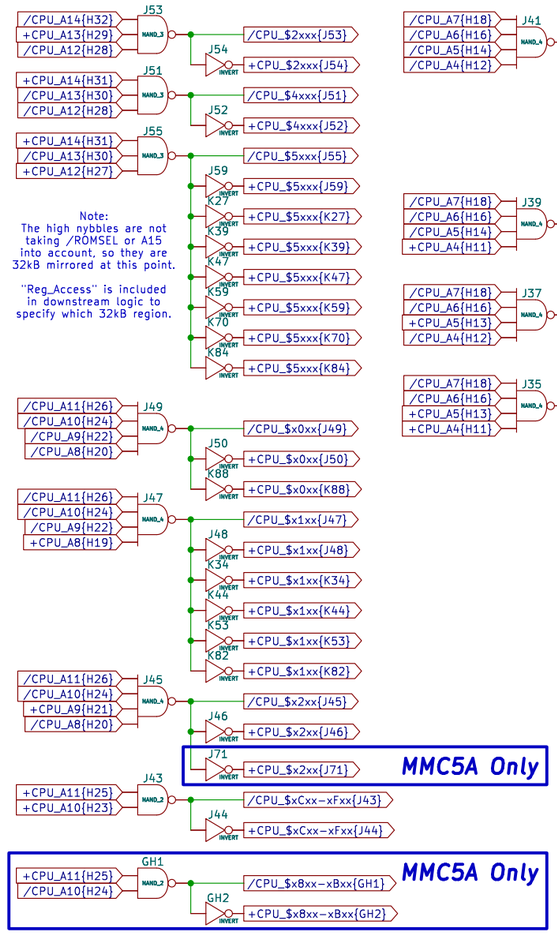
M2 Clock Buffering



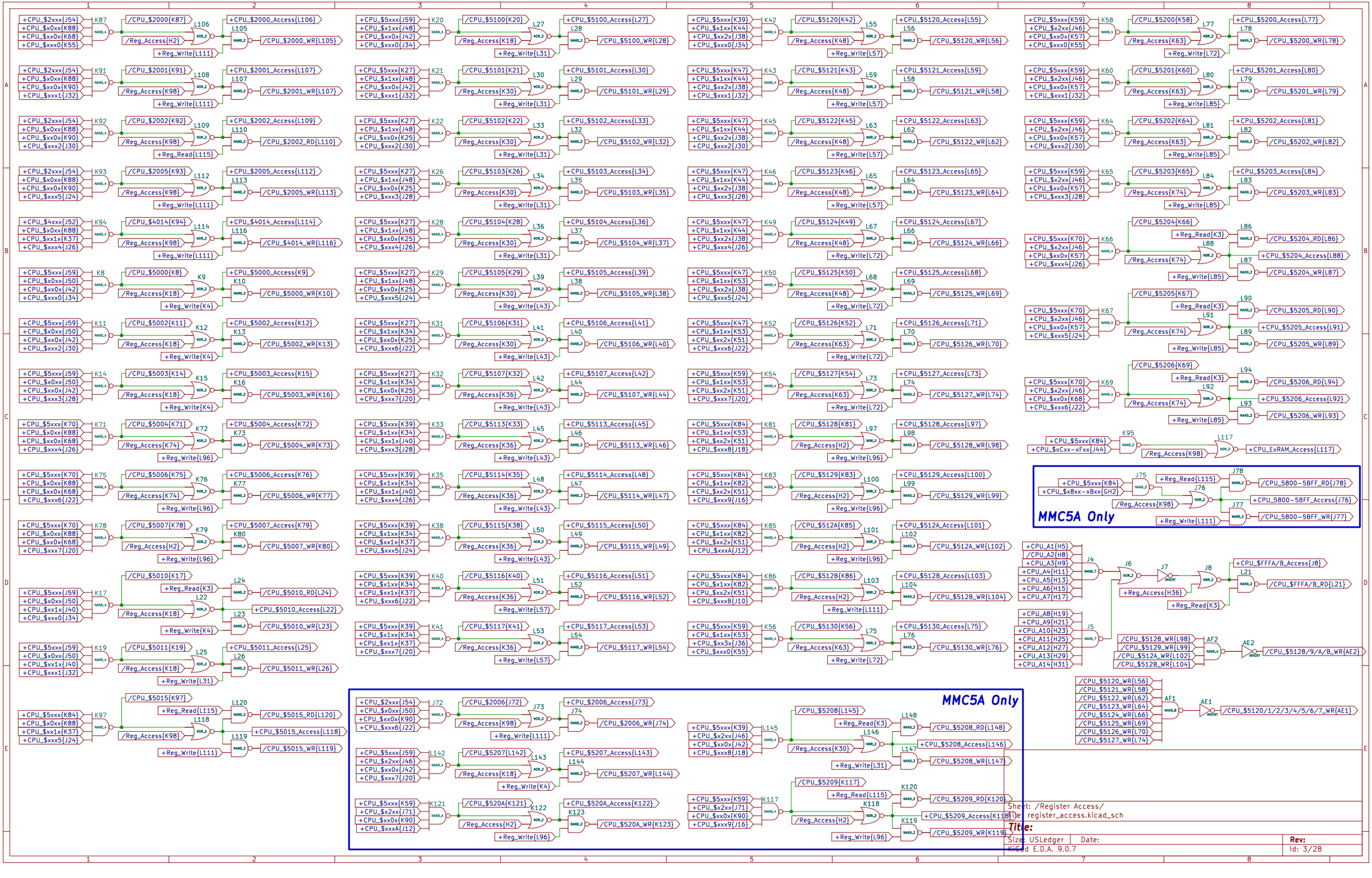
CPU R/W, Register Read/Write



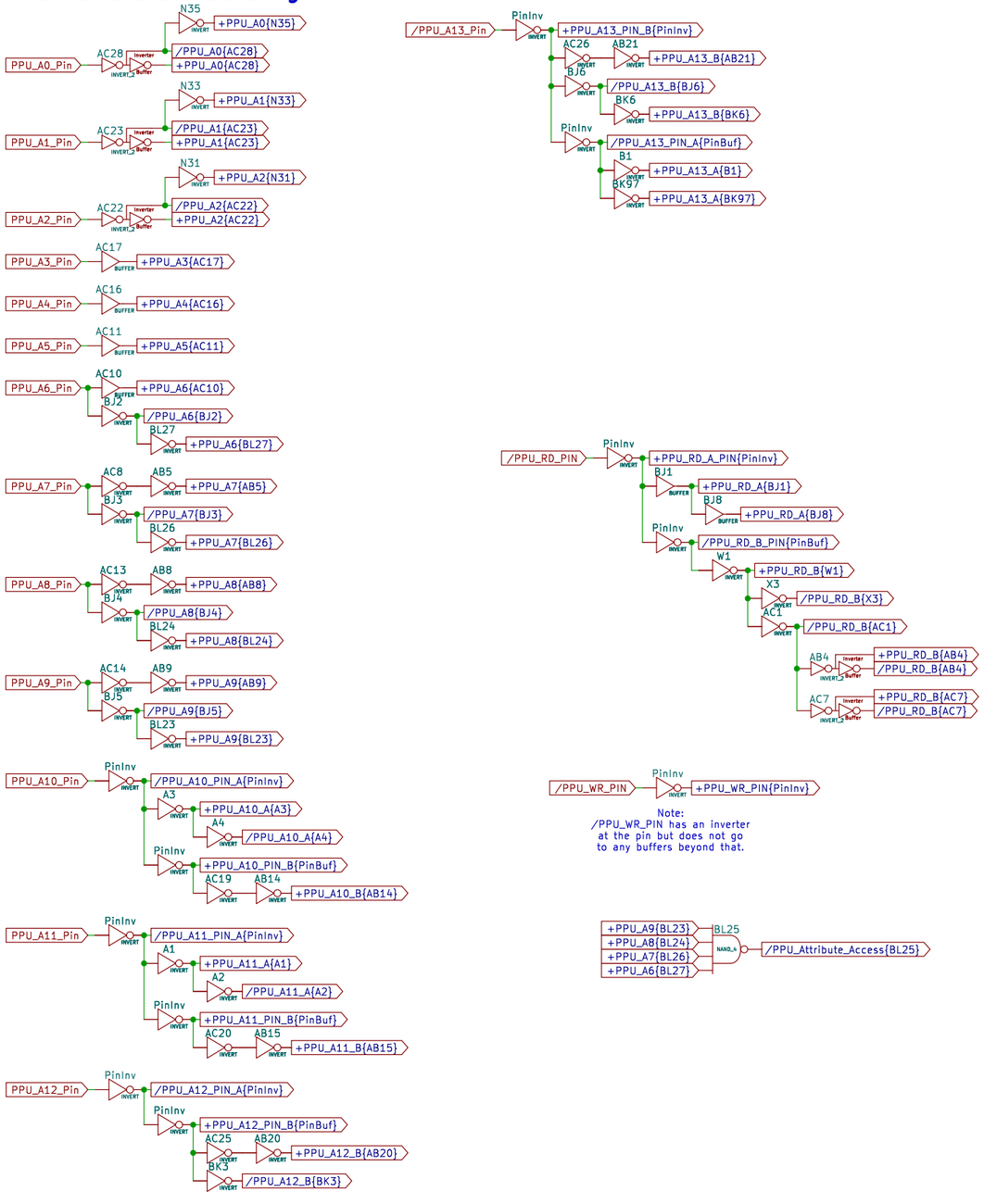
CPU Address Bus Nybble Decoding



Sheet: /CPU Address Bus/	
File: cpuAddressBus.kicad_sch	
Title:	
Size: USLedger	Date:
KiCad E.D.A. 9.0.7	Id: 2/28

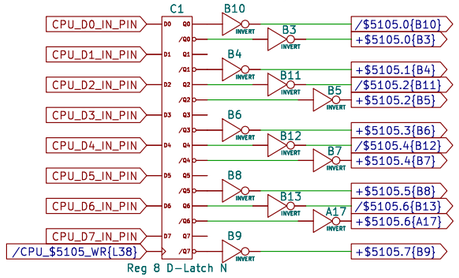

+CPU_$2xxx{J54}
+CPU_$x0xx{K88}
+CPU_$xx0x{K68}
+CPU_$xxx0{K55}
K87
NAND_4
/CPU_$2000{K87}
/Reg_Access{H2}
L106
NOR_2
L105
NAND_2
+CPU_$2000_Access{L106}
/CPU_$2000_WR{L105}
+Reg_Write{L111}
+CPU_$2xxx{J54}
+CPU_$x0xx{K88}
+CPU_$xx0x{K68}
+CPU_$xxx1{J32}
K91
NAND_4
/CPU_$2001{K91}
/Reg_Access{K98}
L108
NOR_2
L107
NAND_2
+CPU_$2001_Access{L107}
/CPU_$2001_WR{L107}
+Reg_Write{L111}
+CPU_$2xxx{J54}
+CPU_$x0xx{K88}
+CPU_$xx0x{K90}
+CPU_$xxx2{J30}
K92
NAND_4
/CPU_$2002{K92}
/Reg_Access{K98}
L109
NOR_2
L110
NAND_2
+CPU_$2002_Access{L109}
/CPU_$2002_RD{L110}
+Reg_Read{L115}
+CPU_$2xxx{J54}
+CPU_$x0xx{K88}
+CPU_$xx0x{K90}
+CPU_$xxx5{J24}
K93
NAND_4
/CPU_$2005{K93}
/Reg_Access{K98}
L112
NOR_2
L113
NAND_2
+CPU_$2005_Access{L112}
/CPU_$2005_WR{L113}
+Reg_Write{L111}
+CPU_$4xxx{J52}
+CPU_$x0xx{K88}
+CPU_$xx1x{K37}
+CPU_$xxx4{J26}
K94
NAND_4
/CPU_$4014{K94}
/Reg_Access{K98}
L114
NOR_2
L116
NAND_2
+CPU_$4014_Access{L114}
/CPU_$4014_WR{L116}
+Reg_Write{L111}
+CPU_$5xxx{J59}
+CPU_$x0xx{J50}
+CPU_$xx0x{J42}
+CPU_$xxx0{J34}
K8
NAND_4
/CPU_$5000{K8}
/Reg_Access{K18}
K9
NOR_2
K10
NAND_2
+CPU_$5000_Access{K9}
/CPU_$5000_WR{K10}
+Reg_Write{K4}
+CPU_$5xxx{J59}
+CPU_$x0xx{J50}
+CPU_$xx0x{J42}
+CPU_$xxx2{J30}
K11
NAND_4
/CPU_$5002{K11}
/Reg_Access{K18}
K12
NOR_2
K13
NAND_2
+CPU_$5002_Access{K12}
/CPU_$5002_WR{K13}
+Reg_Write{K4}
+CPU_$5xxx{J59}
+CPU_$x0xx{J50}
+CPU_$xx0x{J42}
+CPU_$xxx3{J28}
K14
NAND_4
/CPU_$5003{K14}
/Reg_Access{K18}
K15
NOR_2
K16
NAND_2
+CPU_$5003_Access{K15}
/CPU_$5003_WR{K16}
+Reg_Write{K4}
+CPU_$5xxx{K70}
+CPU_$x0xx{K88}
+CPU_$xx0x{K68}
+CPU_$xxx4{J26}
K71
NAND_4
/CPU_$5004{K71}
/Reg_Access{K74}
K72
NOR_2
K73
NAND_2
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/CPU_$5004_WR{K73}
+Reg_Write{L96}
+CPU_$5xxx{K70}
+CPU_$x0xx{K88}
+CPU_$xx0x{K68}
+CPU_$xxx6{J22}
K75
NAND_4
/CPU_$5006{K75}
/Reg_Access{K74}
K76
NOR_2
K77
NAND_2
+CPU_$5006_Access{K76}
/CPU_$5006_WR{K77}
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+CPU_$5xxx{K70}
+CPU_$x0xx{K88}
+CPU_$xx0x{K68}
+CPU_$xxx7{J20}
K78
NAND_4
/CPU_$5007{K78}
/Reg_Access{H2}
K79
NOR_2
K80
NAND_2
+CPU_$5007_Access{K79}
/CPU_$5007_WR{K80}
+Reg_Write{L96}
+CPU_$5xxx{J59}
+CPU_$x0xx{J50}
+CPU_$xx1x{J40}
+CPU_$xxx0{J34}
K17
NAND_4
/CPU_$5010{K17}
/Reg_Access{K18}
+Reg_Read{K3}
L22
NOR_2
L24
NAND_2
L23
NAND_2
/CPU_$5010_RD{L24}
+CPU_$5010_Access{L22}
/CPU_$5010_WR{L23}
+Reg_Write{K4}
+CPU_$5xxx{J59}
+CPU_$x0xx{J50}
+CPU_$xx1x{J40}
+CPU_$xxx1{J32}
K19
NAND_4
/CPU_$5011{K19}
/Reg_Access{K18}
L25
NOR_2
L26
NAND_2
+CPU_$5011_Access{L25}
/CPU_$5011_WR{L26}
+Reg_Write{L31}
+CPU_$5xxx{K84}
+CPU_$x0xx{K88}
+CPU_$xx1x{K37}
+CPU_$xxx5{J24}
K97
NAND_4
/CPU_$5015{K97}
/Reg_Access{K98}
+Reg_Read{L115}
L118
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L120
NAND_2
L119
NAND_2
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+CPU_$5015_Access{L118}
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+Reg_Write{L111}
+CPU_$5xxx{J59}
+CPU_$x0xx{J50}
+CPU_$xx0x{K90}
+CPU_$xxx6{J22}
K20
NAND_4
/CPU_$5100{K20}
/Reg_Access{K18}
L27
NOR_2
L28
NAND_2
+CPU_$5100_Access{L27}
/CPU_$5100_WR{L28}
+Reg_Write{L31}
+CPU_$5xxx{K27}
+CPU_$x1xx{J48}
+CPU_$xx0x{J42}
+CPU_$xxx1{J32}
K21
NAND_4
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/Reg_Access{K30}
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NOR_2
L29
NAND_2
+CPU_$5101_Access{L30}
/CPU_$5101_WR{L29}
+Reg_Write{L31}
+CPU_$5xxx{K27}
+CPU_$x1xx{J48}
+CPU_$xx0x{K25}
+CPU_$xxx2{J30}
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/Reg_Access{K30}
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NOR_2
L32
NAND_2
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+Reg_Write{L31}
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+CPU_$x1xx{J48}
+CPU_$xx0x{K25}
+CPU_$xxx3{J28}
K26
NAND_4
/CPU_$5103{K26}
/Reg_Access{K30}
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NAND_2
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+CPU_$xxx4{J26}
K28
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/Reg_Access{K30}
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NOR_2
L37
NAND_2
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+Reg_Write{L31}
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+CPU_$x1xx{J48}
+CPU_$xx0x{K25}
+CPU_$xxx5{J24}
K29
NAND_4
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/Reg_Access{K30}
L39
NOR_2
L38
NAND_2
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/CPU_$5105_WR{L38}
+Reg_Write{L43}
+CPU_$5xxx{K27}
+CPU_$x1xx{K34}
+CPU_$xx0x{K25}
+CPU_$xxx6{J22}
K31
NAND_4
/CPU_$5106{K31}
/Reg_Access{K30}
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NOR_2
L40
NAND_2
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+Reg_Write{L43}
+CPU_$5xxx{K27}
+CPU_$x1xx{K34}
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+CPU_$xxx7{J20}
K32
NAND_4
/CPU_$5107{K32}
/Reg_Access{K36}
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NOR_2
L44
NAND_2
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+Reg_Write{L43}
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+CPU_$x1xx{K34}
+CPU_$xx1x{J40}
+CPU_$xxx3{J28}
K33
NAND_4
/CPU_$5113{K33}
/Reg_Access{K36}
L45
NOR_2
L46
NAND_2
+CPU_$5113_Access{L45}
/CPU_$5113_WR{L46}
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K35
NAND_4
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/Reg_Access{K36}
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NAND_2
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K38
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NAND_2
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+CPU_$xxx6{J22}
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NOR_2
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NAND_2
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+CPU_$xxx6{J22}
J72
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/CPU_$2006{J72}
/Reg_Access{K98}
J73
NOR_2
J74
NAND_2
+CPU_$2006_Access{J73}
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+CPU_$x2xx{J46}
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NOR_2
L144
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+CPU_$x2xx{J71}
+CPU_$xx0x{K90}
+CPU_$xxxA{J12}
K121
NAND_4
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/Reg_Access{H2}
K122
NOR_2
K123
NAND_2
+CPU_$520A_Access{K122}
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MMC5A Only
+CPU_$5xxx{K39}
+CPU_$x2xx{J46}
+CPU_$xx0x{J42}
+CPU_$xxx8{J18}
L145
NAND_4
/CPU_$5208{L145}
+Reg_Read{K3}
L146
NOR_2
L148
NAND_2
L147
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+Reg_Write{L31}
+CPU_$5xxx{K59}
+CPU_$x2xx{J71}
+CPU_$xx0x{K90}
+CPU_$xxx9{J16}
K117
NAND_4
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K118
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K120
NAND_2
K119
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/Reg_Access{H2}
+Reg_Write{L96}
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K42
NAND_4
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/Reg_Access{K48}
L55
NOR_2
L56
NAND_2
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+Reg_Write{L57}
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+CPU_$x1xx{K44}
+CPU_$xx2x{J38}
+CPU_$xxx1{J32}
K43
NAND_4
/CPU_$5121{K43}
/Reg_Access{K48}
L59
NOR_2
L58
NAND_2
+CPU_$5121_Access{L59}
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+Reg_Write{L57}
+CPU_$5xxx{K47}
+CPU_$x1xx{K44}
+CPU_$xx2x{J38}
+CPU_$xxx2{J30}
K45
NAND_4
/CPU_$5122{K45}
/Reg_Access{K48}
L63
NOR_2
L62
NAND_2
+CPU_$5122_Access{L63}
/CPU_$5122_WR{L62}
+Reg_Write{L57}
+CPU_$5xxx{K47}
+CPU_$x1xx{K44}
+CPU_$xx2x{J38}
+CPU_$xxx3{J28}
K46
NAND_4
/CPU_$5123{K46}
/Reg_Access{K48}
L65
NOR_2
L64
NAND_2
+CPU_$5123_Access{L65}
/CPU_$5123_WR{L64}
+Reg_Write{L57}
+CPU_$5xxx{K47}
+CPU_$x1xx{K44}
+CPU_$xx2x{J38}
+CPU_$xxx4{J26}
K49
NAND_4
/CPU_$5124{K49}
/Reg_Access{K48}
L67
NOR_2
L66
NAND_2
+CPU_$5124_Access{L67}
/CPU_$5124_WR{L66}
+Reg_Write{L72}
+CPU_$5xxx{K47}
+CPU_$x1xx{K53}
+CPU_$xx2x{J38}
+CPU_$xxx5{J24}
K50
NAND_4
/CPU_$5125{K50}
/Reg_Access{K48}
L68
NOR_2
L69
NAND_2
+CPU_$5125_Access{L68}
/CPU_$5125_WR{L69}
+Reg_Write{L72}
+CPU_$5xxx{K47}
+CPU_$x1xx{K53}
+CPU_$xx2x{K51}
+CPU_$xxx6{J22}
K52
NAND_4
/CPU_$5126{K52}
/Reg_Access{K63}
L71
NOR_2
L70
NAND_2
+CPU_$5126_Access{L71}
/CPU_$5126_WR{L70}
+Reg_Write{L72}
+CPU_$5xxx{K59}
+CPU_$x1xx{K53}
+CPU_$xx2x{K51}
+CPU_$xxx7{J20}
K54
NAND_4
/CPU_$5127{K54}
/Reg_Access{K63}
L73
NOR_2
L74
NAND_2
+CPU_$5127_Access{L73}
/CPU_$5127_WR{L74}
+Reg_Write{L72}
+CPU_$5xxx{K84}
+CPU_$x1xx{K53}
+CPU_$xx2x{K51}
+CPU_$xxx8{J18}
K81
NAND_4
/CPU_$5128{K81}
/Reg_Access{H2}
L97
NOR_2
L98
NAND_2
+CPU_$5128_Access{L97}
/CPU_$5128_WR{L98}
+Reg_Write{L96}
+CPU_$5xxx{K84}
+CPU_$x1xx{K82}
+CPU_$xx2x{K51}
+CPU_$xxx9{J16}
K83
NAND_4
/CPU_$5129{K83}
/Reg_Access{H2}
L100
NOR_2
L99
NAND_2
+CPU_$5129_Access{L100}
/CPU_$5129_WR{L99}
+Reg_Write{L96}
+CPU_$5xxx{K84}
+CPU_$x1xx{K82}
+CPU_$xx2x{K51}
+CPU_$xxxA{J12}
K85
NAND_4
/CPU_$512A{K85}
/Reg_Access{H2}
L101
NOR_2
L102
NAND_2
+CPU_$512A_Access{L101}
/CPU_$512A_WR{L102}
+Reg_Write{L96}
+CPU_$5xxx{K84}
+CPU_$x1xx{K82}
+CPU_$xx2x{K51}
+CPU_$xxxB{J10}
K86
NAND_4
/CPU_$512B{K86}
/Reg_Access{H2}
L103
NOR_2
L104
NAND_2
+CPU_$512B_Access{L103}
/CPU_$512B_WR{L104}
+Reg_Write{L111}
+CPU_$5xxx{K59}
+CPU_$x1xx{K53}
+CPU_$xx3x{J36}
+CPU_$xxx0{K55}
K56
NAND_4
/CPU_$5130{K56}
/Reg_Access{K63}
L75
NOR_2
L76
NAND_2
+CPU_$5130_Access{L75}
/CPU_$5130_WR{L76}
+Reg_Write{L72}
+CPU_$5xxx{K59}
+CPU_$x2xx{J46}
+CPU_$xx0x{K57}
+CPU_$xxx0{K55}
K58
NAND_4
/CPU_$5200{K58}
/Reg_Access{K63}
L77
NOR_2
L78
NAND_2
+CPU_$5200_Access{L77}
/CPU_$5200_WR{L78}
+Reg_Write{L72}
+CPU_$5xxx{K59}
+CPU_$x2xx{J46}
+CPU_$xx0x{K57}
+CPU_$xxx1{J32}
K60
NAND_4
/CPU_$5201{K60}
/Reg_Access{K63}
L80
NOR_2
L79
NAND_2
+CPU_$5201_Access{L80}
/CPU_$5201_WR{L79}
+Reg_Write{L85}
+CPU_$5xxx{K59}
+CPU_$x2xx{J46}
+CPU_$xx0x{K57}
+CPU_$xxx2{J30}
K64
NAND_4
/CPU_$5202{K64}
/Reg_Access{K63}
L81
NOR_2
L82
NAND_2
+CPU_$5202_Access{L81}
/CPU_$5202_WR{L82}
+Reg_Write{L85}
+CPU_$5xxx{K59}
+CPU_$x2xx{J46}
+CPU_$xx0x{K57}
+CPU_$xxx3{J28}
K65
NAND_4
/CPU_$5203{K65}
/Reg_Access{K74}
L84
NOR_2
L83
NAND_2
+CPU_$5203_Access{L84}
/CPU_$5203_WR{L83}
+Reg_Write{L85}
+CPU_$5xxx{K70}
+CPU_$x2xx{J46}
+CPU_$xx0x{K57}
+CPU_$xxx4{J26}
K66
NAND_4
/CPU_$5204{K66}
+Reg_Read{K3}
L88
NOR_2
L86
NAND_2
L87
NAND_2
/CPU_$5204_RD{L86}
+CPU_$5204_Access{L88}
/CPU_$5204_WR{L87}
/Reg_Access{K74}
+Reg_Write{L85}
+CPU_$5xxx{K70}
+CPU_$x2xx{J46}
+CPU_$xx0x{K57}
+CPU_$xxx5{J24}
K67
NAND_4
/CPU_$5205{K67}
+Reg_Read{K3}
L91
NOR_2
L90
NAND_2
L89
NAND_2
/CPU_$5205_RD{L90}
+CPU_$5205_Access{L91}
/CPU_$5205_WR{L89}
/Reg_Access{K74}
+Reg_Write{L85}
+CPU_$5xxx{K70}
+CPU_$x2xx{J46}
+CPU_$xx0x{K68}
+CPU_$xxx6{J22}
K69
NAND_4
/CPU_$5206{K69}
+Reg_Read{K3}
L92
NOR_2
L94
NAND_2
L93
NAND_2
/CPU_$5206_RD{L94}
+CPU_$5206_Access{L92}
/CPU_$5206_WR{L93}
/Reg_Access{K74}
+Reg_Write{L85}
+CPU_$5xxx{K84}
+CPU_$xCxx-xFxx{J44}
K95
NAND_2
L117
NOR_2
+CPU_ExRAM_Access{L117}
/Reg_Access{K98}
MMC5A Only
+CPU_$5xxx{K84}
+CPU_$x8xx-xBxx{GH2}
J75
NAND_2
+Reg_Read{L115}
J78
NAND_2
/CPU_5800-5BFF_RD{J78}
J76
NOR_2
+CPU_5800-5BFF_Access{J76}
/Reg_Access{K98}
J77
NAND_2
+Reg_Write{L111}
/CPU_5800-5BFF_WR{J77}
+CPU_A1{H5}
/CPU_A2{H8}
+CPU_A3{H9}
+CPU_A4{H11}
+CPU_A5{H13}
+CPU_A6{H15}
+CPU_A7{H17}
J4
NAND_7
J6
NOR_2
J7
INVERT
J8
NOR_2
+CPU_$FFFA/B_Access{J8}
L21
NAND_2
/CPU_$FFFA/B_RD{L21}
+Reg_Access{H36}
+Reg_Read{K3}
+CPU_A8{H19}
+CPU_A9{H21}
+CPU_A10{H23}
+CPU_A11{H25}
+CPU_A12{H27}
+CPU_A13{H29}
+CPU_A14{H31}
J5
NAND_7
/CPU_$5128_WR{L98}
/CPU_$5129_WR{L99}
/CPU_$512A_WR{L102}
/CPU_$512B_WR{L104}
AF2
NAND_4
AE2
INVERT
/CPU_$5128/9/A/B_WR{AE2}
/CPU_$5120_WR{L56}
/CPU_$5121_WR{L58}
/CPU_$5122_WR{L62}
/CPU_$5123_WR{L64}
/CPU_$5124_WR{L66}
/CPU_$5125_WR{L69}
/CPU_$5126_WR{L70}
/CPU_$5127_WR{L74}
AF1
NAND_8
AE1
INVERT
/CPU_$5120/1/2/3/4/5/6/7_WR{AE1}
Sheet: /Register Access/
File: register_access.kicad_sch
Title:
Size: USLedger
Date:
KiCad E.D.A. 9.0.7
Rev:
Id: 3/28

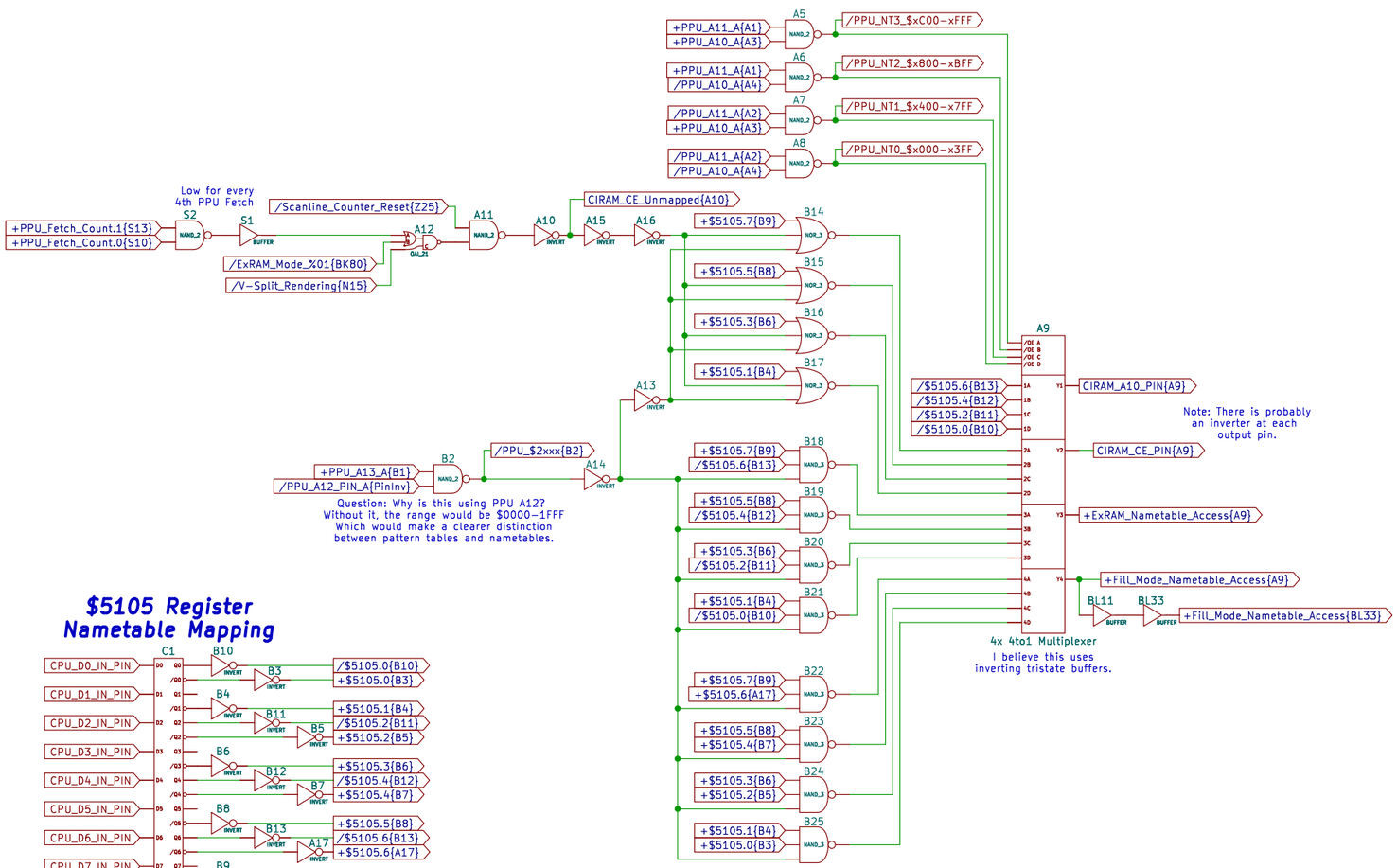
PPU Address Bus Buffering



Sheet: /PPU Address Bus/	
File: ppu_address_bus.kicad_sch	
Title:	
Size: USLedger Date:	**Rev:**
KiCad E.D.A. 9.0.7	Id: 4/28

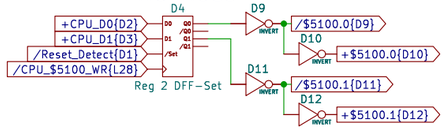
$5105 Register Nametable Mapping



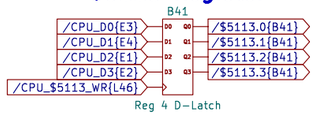


Sheet: /Nametable Mapping/		
File: nametable_mapping.kicad_sch		
Title:		
Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 5/28

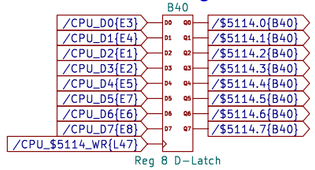
$5100 Register
PRG Mode



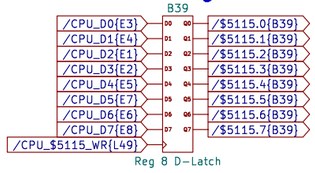
PRG Bank Registers
$5113 Register



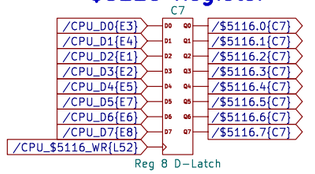
$5114 Register



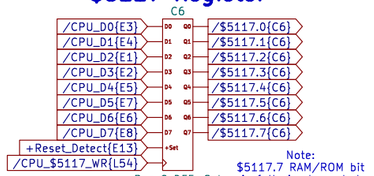
$5115 Register



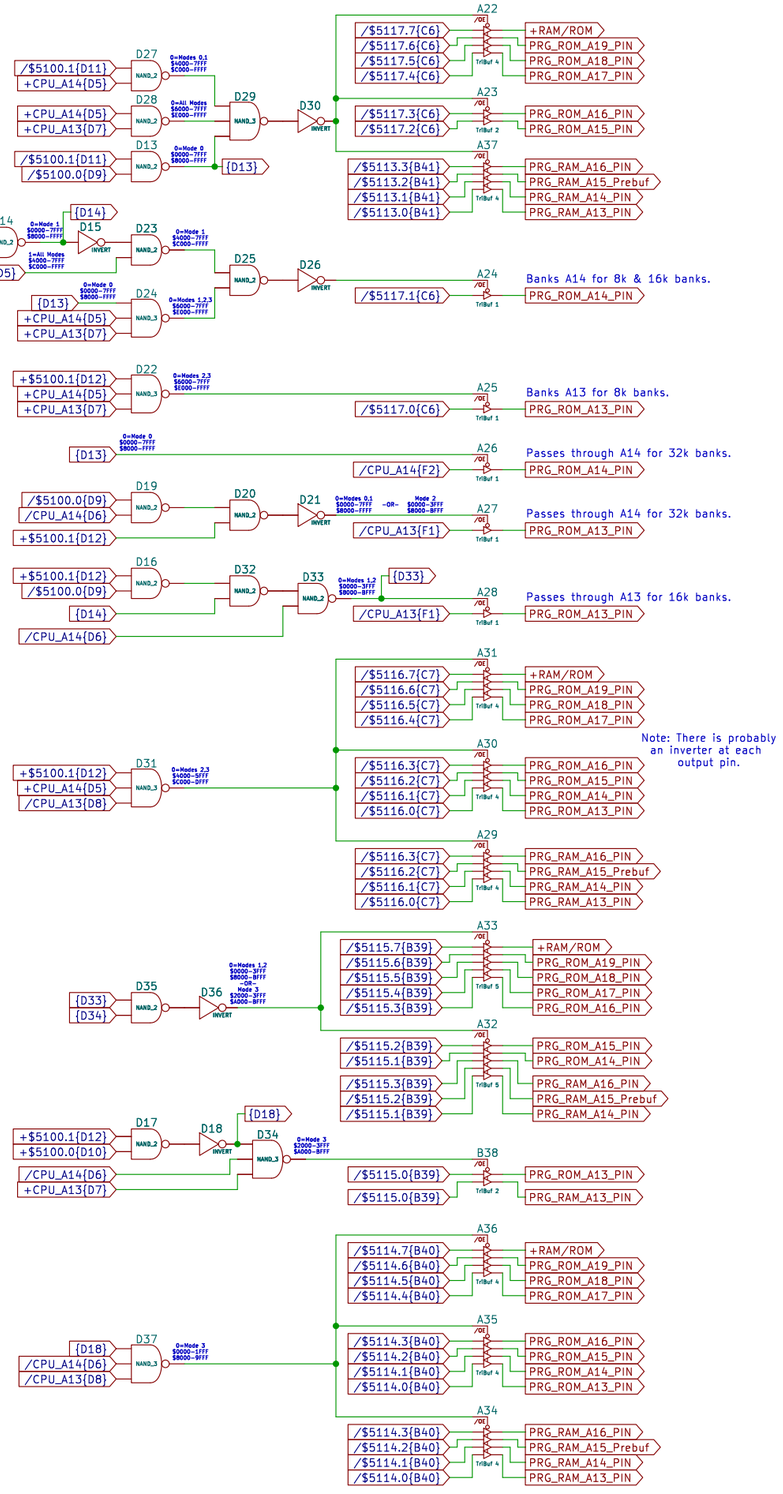
$5116 Register

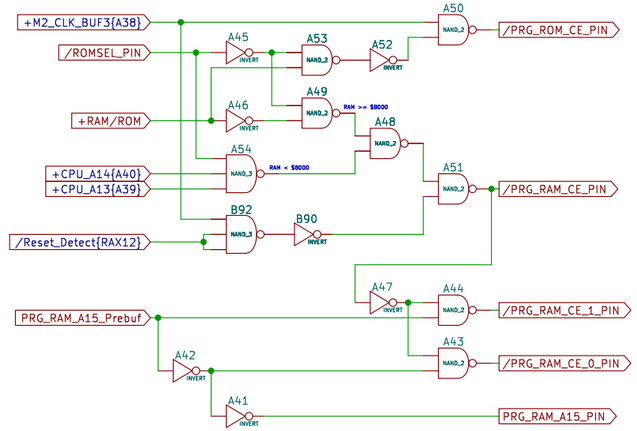


$5117 Register

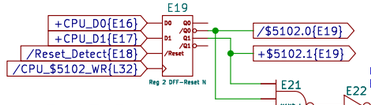


Note: $5117.7 RAM/ROM bit is fully implemented but is somehow being prevented from entering RAM mode when bench testing.

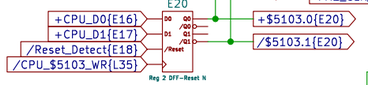




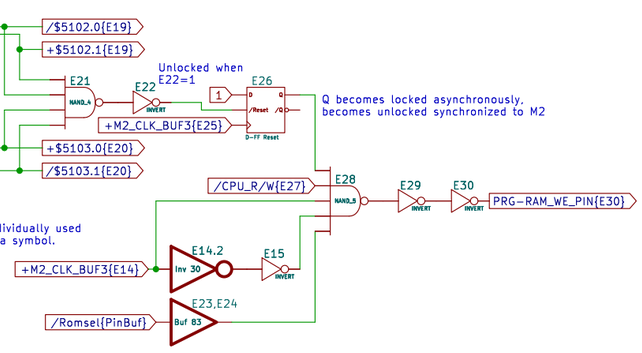
$5102 Register
PRG-RAM Write Protection



$5103 Register



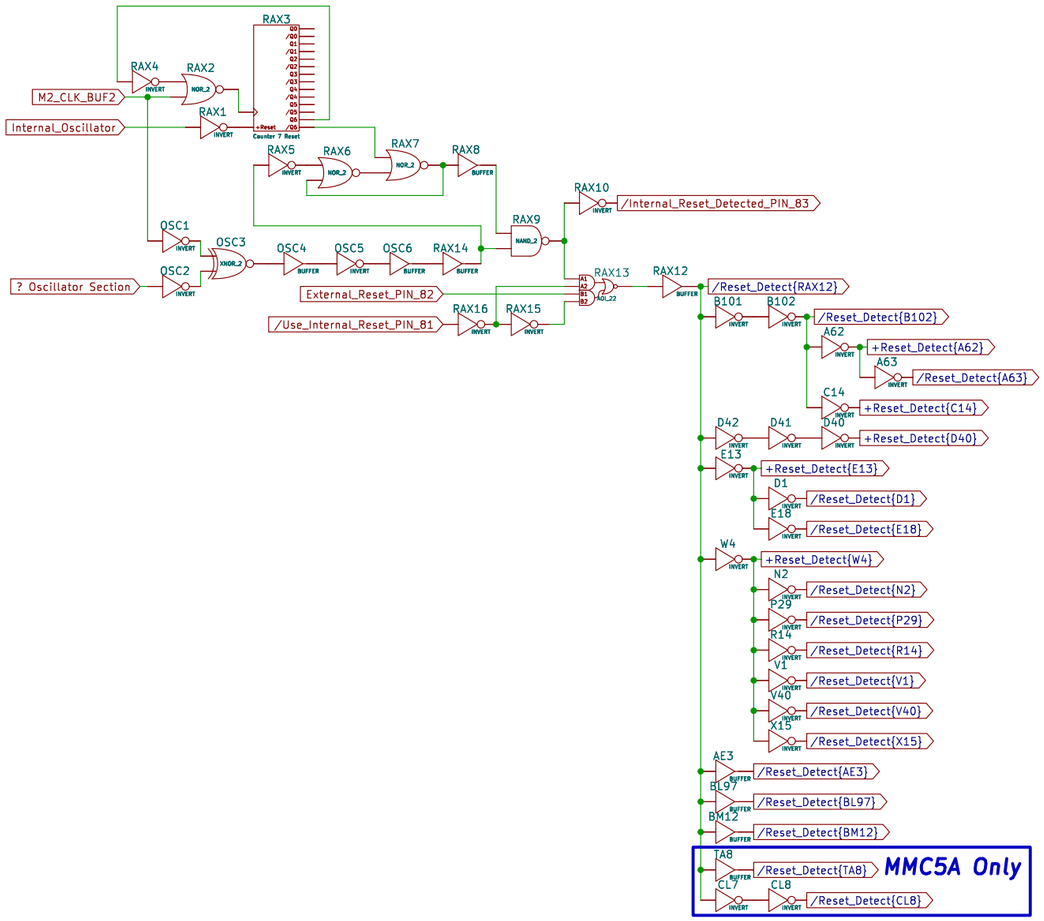
Note: /Set vs. /Reset inputs are individually used but not easily shown here with a symbol.

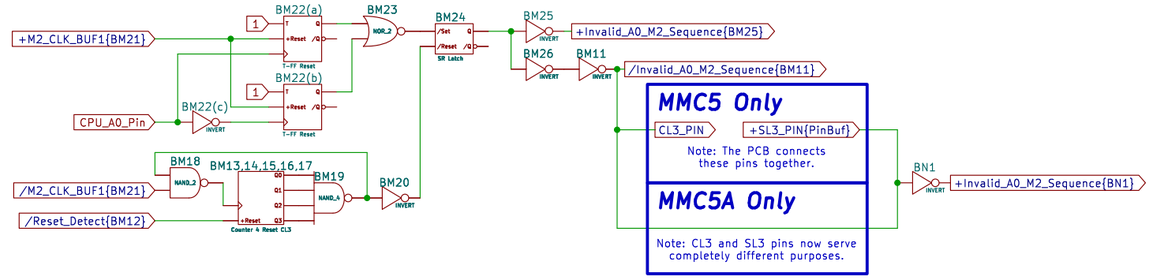


Sheet: /PRG Banking/
File: prg_banking.kicad_sch

Title:

Size: USLedger	Date:	Rev:
KiCad E.D.A. 9.0.7		Id: 6/28





Sheet: /Reset Detection/
File: reset_detection.kicad_sch

Title:

Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 7/28

Incoming CPU Data Bus Buffering Bonanza

CPU_D0_IN_PIN → E3 → /CPU_D0{E3}
- D2 → +CPU_D0{D2}
- E16 → +CPU_D0{E16}
- G15 → G16 → +CPU_D0{G16}
- L122 → /CPU_D0{L122}
- X18 → /CPU_D0{X18}
- P26 → +CPU_D0{P26}
- Y4 → +CPU_D0{Y4}
- AE16 → /CPU_D0{AE16}
- AE18 → /CPU_D0{AE18}
- MUL19 → +CPU_D0{MUL19}
- TF1 → +CPU_D0{TF1}
- CL5 → +CPU_D0{CL5}

MMC5A Only

CPU_D1_IN_PIN → E4 → /CPU_D1{E4}
- D3 → +CPU_D1{D3}
- E17 → +CPU_D1{E17}
- G13 → G14 → +CPU_D1{G14}
- L123 → /CPU_D1{L123}
- X19 → /CPU_D1{X19}
- P19 → +CPU_D1{P19}
- Y5 → +CPU_D1{Y5}
- AE15 → /CPU_D1{AE15}
- AE19 → /CPU_D1{AE19}
- MUL20 → +CPU_D1{MUL20}
- TF2 → +CPU_D1{TF2}
- CL6 → +CPU_D1{CL6}

MMC5A Only

CPU_D2_IN_PIN → E1 → /CPU_D2{E1}
- G11 → G12 → +CPU_D2{G12}
- L124 → /CPU_D2{L124}
- X20 → /CPU_D2{X20}
- P13 → +CPU_D2{P13}
- Y11 → +CPU_D2{Y11}
- AE20 → /CPU_D2{AE20}
- MUL21 → +CPU_D2{MUL21}
- TF3 → +CPU_D2{TF3}

MMC5A Only

CPU_D3_IN_PIN → E2 → /CPU_D3{E2}
- G9 → G10 → +CPU_D3{G10}
- E55 → /CPU_D3{E55}
- E60 → +CPU_D3{E60}
- L125 → /CPU_D3{L125}
- X21 → /CPU_D3{X21}
- Y12 → +CPU_D3{Y12}
- AE25 → /CPU_D3{AE25}
- MUL22 → +CPU_D3{MUL22}
- TF4 → +CPU_D3{TF4}

MMC5A Only

CPU_D4_IN_PIN → E5 → /CPU_D4{E5}
- X22 → /CPU_D4{X22}
- Y18 → +CPU_D4{Y18}
- G7 → G8 → +CPU_D4{G8}
- E56 → /CPU_D4{E56}
- E61 → +CPU_D4{E61}
- L126 → /CPU_D4{L126}
- AE24 → /CPU_D4{AE24}
- MUL23 → +CPU_D4{MUL23}
- TF5 → +CPU_D4{TF5}

MMC5A Only

CPU_D5_IN_PIN → E7 → /CPU_D5{E7}
- G5 → G6 → +CPU_D5{G6}
- E57 → /CPU_D5{E57}
- E62 → +CPU_D5{E62}
- L127 → /CPU_D5{L127}
- X23 → /CPU_D5{X23}
- Y19 → +CPU_D5{Y19}
- AE23 → /CPU_D5{AE23}
- MUL24 → +CPU_D5{MUL24}
- TF6 → +CPU_D5{TF6}

MMC5A Only

CPU_D6_IN_PIN → E6 → /CPU_D6{E6}
- G3 → G4 → +CPU_D6{G4}
- E58 → /CPU_D6{E58}
- E63 → +CPU_D6{E63}
- L128 → /CPU_D6{L128}
- X24 → /CPU_D6{X24}
- P7 → +CPU_D6{P7}
- Y25 → +CPU_D6{Y25}
- AE22 → /CPU_D6{AE22}
- MUL25 → +CPU_D6{MUL25}
- TF7 → +CPU_D6{TF7}
- CL4 → +CPU_D6{CL4}

MMC5A Only

CPU_D7_IN_PIN → E8 → /CPU_D7{E8}
- G1 → G2 → +CPU_D7{G2}
- E59 → /CPU_D7{E59}
- E64 → +CPU_D7{E64}
- L129 → /CPU_D7{L129}
- X25 → /CPU_D7{X25}
- P5 → +CPU_D7{P5}
- Y26 → +CPU_D7{Y26}
- AE21 → /CPU_D7{AE21}
- MUL26 → +CPU_D7{MUL26}
- TF8 → +CPU_D7{TF8}
- CL3 → +CPU_D7{CL3}

MMC5A Only

Status Bit Buffers

$5015.0_Read_Pulse_1_Status_Bit{B106}	BL21	$5015.0_Read_Pulse_1_Status_Bit{BL21}
$5015.1_Read_Pulse_2_Status_Bit{B104}	BL20	$5015.1_Read_Pulse_2_Status_Bit{BL20}
+Invalid_A0_M2_Sequence{BM25}	BL52	$5204.0_Read_Invalid_A0_M2_Sequence{BL52}
+PCM_IRQ_Status_Bit{K114}	BL62	$5010.7_Read_PCM_IRQ_Pending{BL62}
/Scanline_IRQ_Status{X28}	BL79	$5204.7_Read_Scanline_IRQ_Pending{BL79}
+In_Frame{Z21}	BL80	$5010.6_Read_In_Frame{BL80}
Pin_97_CL3_GPIO_In	BM0.7	$5208.6_CL3_GPIO_In{BM0.7}
Pin_98_SL3_GPIO_In	BM0.6	$5208.7_SL3_GPIO_In{BM0.6}
+Timer_IRQ_Status_Bit{TA22}	BM0.4	$5209.7_Timer_IRQ_Status_Bit{BM0.4}

MMC5A Only

Decides When to Drive the CPU Data Bus

MMC5A Only

+CPU_$5208_Access{L146}	BM0.2	BP0
+CPU_$5209_Access{K118}	BM0.1	BK54.5A

BP2 NAND_6

MMC5 Only

+CPU_$5015_Access{L118}	BL22	+CPU_$5015_Access{BL22}	BP1
+CPU_$5204_Access{L88}	BL56	+CPU_$5204_Access{BL56}	BK48
+CPU_$5205_Access{L91}	BL55	+CPU_$5205_Access{BL55}	BP3
+CPU_$5206_Access{L92}	BL54	+CPU_$5206_Access{BL54}	BP4
+CPU_ExRAM_Access{L117}	BL51	+CPU_ExRAM_Access{BL51}	BK52
+ExRAM_Mode_$0x{BL96}	BL48	+ExRAM_Mode_$0x{BL48}	BK50
+CPU_$5010_Access{L22}	BL53	+CPU_$5010_Access{BL53}	BK49
/Reg_Read{K2}	BL50		BK47

BP2 NAND_4 → BK51 → BK46 → BK45 → CPU_Dx_Direction{BK45}

Multiplexing Data to be Driven onto the CPU Data Bus

I wonder why they used _RD instead of _Access for the new status bits? They even went so far as to run $5010_RD way across the die when $5010_Access was already here.

CPU_Dx_Direction{BK45} already considers Reg_Read and its applicable /ROMSEL delay.

MMC5A Only

/CPU_$5010_RD{L24}	BM0.8	BN0.6
/CPU_$5208_RD{L148}	BM0.5	BN0.3
/CPU_$5209_RD{K120}	BM0.3	BN0.1

+CPU_$5206_Access{BL54} → BL58
+CPU_$5205_Access{BL55} → BL59
+CPU_$5204_Access{BL56} → BL60
+CPU_$5010_Access{BL53} → BL57
+CPU_$5015_Access{BL22}
+CPU_ExRAM_Access{BL51}
+ExRAM_Mode_$0x{BL48}

BL61.1 INVERT → BL61.2 NOR_2

$5010.0 MMC5A Revision Bit = 1

ExRAM_D0_Read{BJ12(a)}	BK53	BK55	CPU_D0_OUT_PIN
ExRAM_D1_Read{BJ12(b)}	BK58	BK56	CPU_D1_OUT_PIN
ExRAM_D2_Read{BJ12(c)}	BK59	BK61	CPU_D2_OUT_PIN
ExRAM_D3_Read{BJ12(d)}	BK64	BK62	CPU_D3_OUT_PIN
ExRAM_D4_Read{BJ12(e)}	BK65	BK67	CPU_D4_OUT_PIN
ExRAM_D5_Read{BJ12(f)}	BK70	BK68	CPU_D5_OUT_PIN
ExRAM_D6_Read{BJ12(g)}	BK71	BK73	CPU_D6_OUT_PIN
ExRAM_D7_Read{BJ12(h)}	BK76	BK74	CPU_D7_OUT_PIN

$5206.0_Read_Multiply_Result_D8{BL78}	$5206.1_Read_Multiply_Result_D9{BL77}	$5206.2_Read_Multiply_Result_D10{BL76}
$5204.0_Read_Invalid_A0_M2_Sequence{BL52}	$5205.1_Read_Multiply_Result_D1{BL69}	$5205.2_Read_Multiply_Result_D2{BL68}
$5205.0_Read_Multiply_Result_D0{BL70}	$5015.1_Read_Pulse_2_Status_Bit{BL20}	$5206.3_Read_Multiply_Result_D11{BL75}
$5015.0_Read_Pulse_1_Status_Bit{BL21}		$5205.3_Read_Multiply_Result_D3{BL67}
		$5206.4_Read_Multiply_Result_D12{BL74}
		$5205.4_Read_Multiply_Result_D4{BL66}
		$5206.5_Read_Multiply_Result_D13{BL73}
		$5205.5_Read_Multiply_Result_D5{BL65}
		$5206.6_Read_Multiply_Result_D14{BL64}
		$5205.6_Read_Multiply_Result_D6{BL72}
		$5010.6_Read_In_Frame{BL80}
		$5206.7_Read_Multiply_Result_D15{BL63}
		$5205.7_Read_Multiply_Result_D7{BL71}
		$5204.7_Read_Scanline_IRQ_Pending{BL79}
		$5010.7_Read_PCM_IRQ_Pending{BL62}

MMC5 Only

MMC5A Only

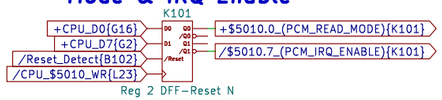
$5208.6_CL3_GPIO_In{BM0.7}	BN0.4	BR2
$5209.7_Timer_IRQ_Status_Bit{BM0.4}	BN0.2	BR1
$5208.7_SL3_GPIO_In{BM0.6}		

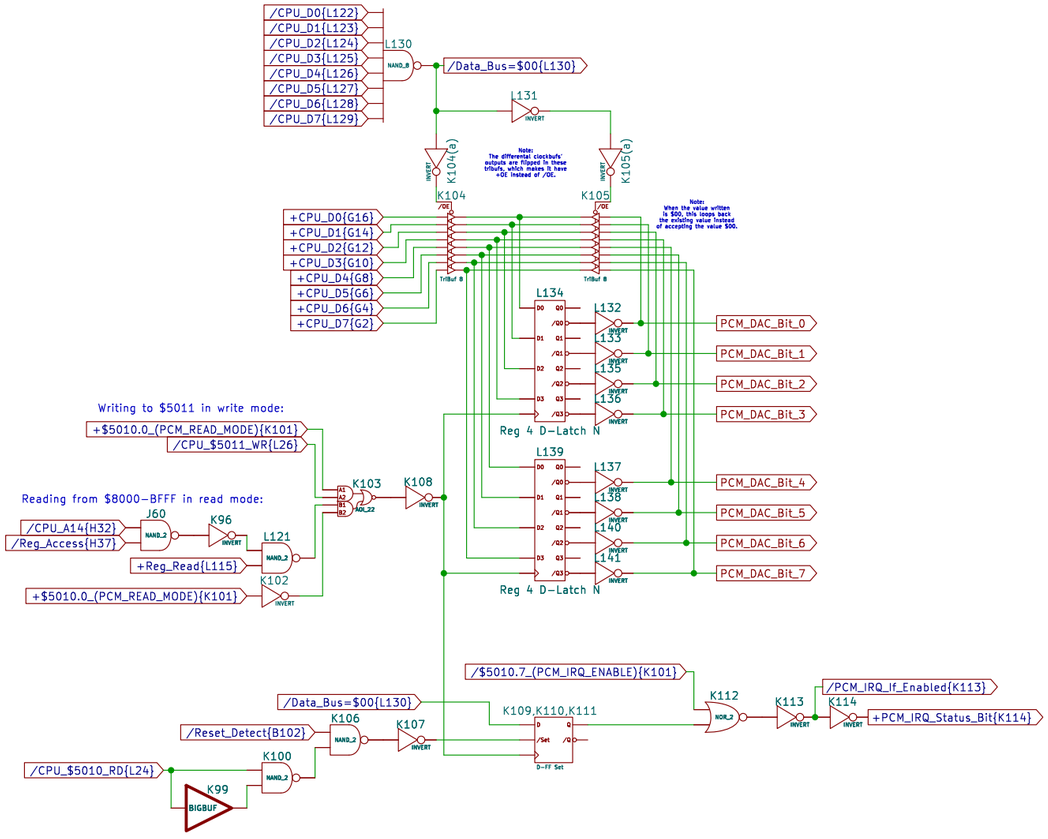
BN0.5 → BR3

MMC5A Only

Sheet:	/CPU Data Bus/
File:	cpu_data_bus.kicad_sch
Title:	
Size:	USLedger
Date:	
Rev:	
KiCad E.D.A.	9.0.7
Id:	8/28

$5010 Register
Mode & IRQ Enable





Sheet: /PCM DAC/	
File: pcm_dac.kicad_sch	
Title:	
Size: USLedger | Date:	**Rev:**
KiCad E.D.A. 9.0.7	Id: 9/28

Scanline Start Detection

+Use Internal Scanline Detection

+Use External Scanline Detection

+Test_Mode_PIN_92{Pin1nv} → AA1 (Inverter, Buffer)

/Test_Input_PIN_29{Pin1nv} → AA4 (INVERT)

External Scanline Start

High when the present PPU address is from $2xxx and matches the previous read.

AB2 (SMR Reg 3) AB2(a) (INVERT) ← +PPU_RD_B{W1}

AB1 (NAND_4)

Low when the 3 most recent PPU reads were from $2xxx and all matched.

AA5 (NOR_4) ← AA1, AB1

AA2 (AOI_22) ← AA1, AA4, AB1, AB2

AA3 (INVERT) → /Scanline_Start_Detected{AA3}

R2 (BUFFER) → /Scanline_Start_Detected{R2}

AC2 (Counter 2 Reset) ← M2_CLK_BUF1, /PPU_RD_B{AC1}

AC6 (NAND_2) ← AC2 → /Exit_In_Frame{AC6}

Scanline End Detection

Increment_Scanline_Fetch_Count{S6}, +PPU_Cycle_336{R4} → R6 (NAND_2) → R7 (INVERT) → R8 (INVERT) → Scanline_End_Detected_Internal{R8}

Scanline_End_Detected_Internal{R8} → R1 (AOI_22)

/Test_Input_PIN_30{Pin1nv} → P4 (INVERT)

External Scanline End

R1 → P3 (INVERT) → Scanline_End_Detected{P3}

P3 → T5 (BUFFER) → Scanline_End_Detected{T5}

$5204 Register Scanline IRQ Enable

+CPU_D7{Y26}, /Reset_Detect{X15}, /CPU_$5204_WR{L87} → X12 (D-FF Reset)

X12 → X13 (NAND_2) → X14 (BUFFER) → +Scanline_IRQ_Pin{X14}

This signal likely needs to be active low.

$5203 Register IRQ Scanline Compare Value

Y2,7,9,14,16,21,23,28

+CPU_D0{Y4}, +CPU_D1{Y5}, +CPU_D2{Y11}, +CPU_D3{Y12}, +CPU_D4{Y18}, +CPU_D5{Y19}, +CPU_D6{Y25}, +CPU_D7{Y26} → Y1 (BUFFER)

/CPU_$5203_WR{L83} → Y1

Reg 8 D-Latch

Y3, Y6, Y10, Y13, Y17, Y20, Y24, Y27 (XNOR_2) → Y8 (NAND_4), Y22 (NAND_4) → Y15 (NOR_2)

Hi when $5203 matches the present V-Split scanline.

D'Morgan tells us that we can pretend Y8,Y15,Y22 are all AND gates.

IRQ Scanline Counter

Z2–Z18 (Counter 8 Reset)

/Scanline_Start_Detected{AA3}, +In_Frame{Z21} → Z1 (NAND_2)

/In_Frame{Z23} → Z19 (BUFFER)

Z1, Z19 → Z22 (NAND_4) → /Scanline_240_and_up{Z22}

$2001 Register Write Snooping

+CPU_D3{Y12}, +CPU_D4{Y18}, /CPU_$2001_WR{L107} → X8 (Reg 2 DFF-No-Reset N)

X8 → X9 (NAND_2)

Hi = PPU Rendering Enabled

$2002 Register Read Snooping

/CPU_$2002_RD{L110} → X16 (INVERT) → X4 (NAND_2)

+CPU_D7{Y26}, M2_CLK_BUF1 → X5 (D-FF No Reset)

Latches $2002.7 value read by CPU for 1 CPU cycle.

/CPU_$FFFA/B_RD{L21}, /CPU_$4014_WR{L116}, /Exit_In_Frame{AC6} → X6 (NAND_4) → X7 (INVERT)

/Scanline_Start_Detected{AA3} → Z23 (D-FF Reset)

Z23 → /In_Frame{Z23}

Z23 → Z21 (INVERT) → +In_Frame{Z21}

/In_Frame{Z23} → Z24 (NAND_2) → Z25 (INVERT) → +Rendering_In_Frame{Z25}

V39 (INVERT) → /Rendering_In_Frame{V39}

R24 (INVERT) → /Rendering_In_Frame{R24}

N16 (INVERT) → N9 (INVERT) → +Rendering_In_Frame{N9}

Frame Detection

/CPU_$5204_RD{L86} → M3 (BIGBUF) → N38 (NAND_2) → N39 (INVERT) → M2 (NAND_2)

+New_Scanline_Detected{AA3 via Z23}, /In_Frame{Z23} → Z20 (NAND_2) → X17 (NAND_2) → X26 (INVERT)

/Scanline_Start_Detected{AA3} → Y31 (D-FF Reset)

Y31 → Y29 (NOR_2) → Y30 (INVERT) → /CPU_$5204_RD{L86}

+IRQ

Y31 → Y32 (INVERT) → X27 (D-FF No Reset) → X28 (INVERT) → /Scanline_IRQ_Status{X28}

/IRQ

This signal likely needs to be active hi.

Address Matching

/PPU_RD_B{AB4} → AA7 (INVERT)

/Exit_In_Frame{AC6} → AA6 (INVERT)

/PPU_RD_B{AC7} → AB3 (INVERT)

When not in frame, the previous PPU read address always looks like $0000. Since A13, A12 are looking at range $2xxx instead of matches, $0xxx won't match $2xxx. Therefore, scanlines don't get detected when using PPUDATA to read from $0000 during v-blank.

+PPU_A13_B{AB21}	AB22	AA21 (NAND_2)	AA20 (INVERT)
+PPU_A12_B{AB20}	AB19	AA18 (NOR_2)	
+PPU_A11_B{AB15}	AB16	AA17 (XNOR_2)	
+PPU_A10_B{AB14}	AB13	AA14 (XNOR_2)	
+PPU_A9{AB9}	AB10	AA13 (XNOR_2)	
+PPU_A8{AB8}	AB7	AA12 (XNOR_2)	
+PPU_A7{AB5}	AB6	AA9 (XNOR_2)	
+PPU_A6{AC10}	AC9	AA8 (XNOR_2)	
+PPU_A5{AC11}	AC12	AA11 (XNOR_2)	
+PPU_A4{AC16}	AC15	AB11 (XNOR_2)	
+PPU_A3{AC17}	AC18	AB12 (XNOR_2)	
+PPU_A2{AC22}	AC21	AB17 (XNOR_2)	
+PPU_A1{AC23}	AC24	AB18 (XNOR_2)	
+PPU_A0{AC28}	AC27	AA22 (XNOR_2)	

All D-FF Reset

AA16 (NAND_4) ← AA21, AA18, AA17, AA14, AA13, AA12, AA9, AA8

AA10 (NAND_4) ← AA16, AA5

AA15 (NAND_4) ← AA11, AA5

AA19 (NAND_2) ← AB18, AA15

V−Split Scanline Counter

$5201 Register V−Split V−Scroll

W3,V24,V25,V41

/CPU_D0{X18}
/CPU_D1{X19}
/CPU_D2{X20}
/CPU_D3{X21}
/CPU_D4{X22}
/CPU_D5{X23}
/CPU_D6{X24}
/CPU_D7{X25}
/Reset_Detect{V40}
/CPU_$5201_WR{L79}

Reg 8 DFF−Reset

Scanline_End_Detected{P3}

/Rendering_In_Frame{V39}
Reload_V−Split_Scanline_Count{T1}

V15 V14 V13 T7 T8 T9 +V−Split_Scanline_Counter.0{T9}
V18 V17 V16 T10 T12 T13 +V−Split_Scanline_Counter.1{T13}
V21 V20 V19 T14 T15 T16 +V−Split_Scanline_Counter.2{T16}
V26 V23 V22 T17 T18 T19 +V−Split_Scanline_Counter.3{T19}
V29 V28 V27 T20 T21 T22 +V−Split_Scanline_Counter.4{T22}
V32 V31 V30 T23 T24 T25 +V−Split_Scanline_Counter.5{T25}
V35 V34 V33 T26 T27 T28 +V−Split_Scanline_Counter.6{T28}
V38 V37 V36 T29 T30 T31 +V−Split_Scanline_Counter.7{T31}

NAND_2, INVERT, T−FF Preload

$5200 Register V−Split Mode & H−Threshold

W2

/CPU_D0{X18}
/CPU_D1{X19}
/CPU_D2{X20}
/CPU_D3{X21}
/CPU_D4{X22}
/Reset_Detect{V1}
/CPU_$5200_WR{L78}

V2 BUFFER Reg 5 D−Latch−Reset

Note: Bit 5 does not exist.

+PPU_Fetch_Counter.2{S18}
+PPU_Fetch_Counter.3{S21}
+PPU_Fetch_Counter.4{S24}
+PPU_Fetch_Counter.5{S25}
+PPU_Fetch_Counter.6{S17}
+PPU_Fetch_Counter.7{S29}

V3 V5 V6 V10 V12 XNOR_2
V4 V11 NAND_3
T6 INVERT
V9 NOR_2
V8 D−FF No Reset
V7 INVERT

Comparing the fetch counter, shifted right by 2, to the $5200 V−Split H−threshold, Every 4 fetches = 1 tile. The match at V9..V7 holds true only for 1 fetch, so it is latched by N3/N4 until a new scanline is detected.

XNor N5 selects the left/right side based on $5002.6

+$5200.6_V−Split_Side_Right{P6}
N5 XNOR_2 V−Split_Rendering{N5}
N3 N4 NAND_2
Scanline_End_Detected{P3}

P6

+CPU_D6{P7}
+CPU_D7{P5}
/Reset_Detect{N2}
/CPU_$5200_WR{L78}

N1 BUFFER Reg 2 DFF−Reset N

+$5200.6_V−Split_Side_Right{P6}
+$5200.7_V−Split_Enable{P6}

/Any_PPU_Fetch_While_In_Frame{X1}

+PPU_Fetch_Counter.0{S10}
+PPU_Fetch_Counter.1{S13}
+PPU_Fetch_Counter.2{S18}
/PPU_Fetch_Counter.4{S23}
+PPU_Fetch_Counter.7{S29}

S3 NAND_5

S3 = 0 at these values:
135, 143 (cycle 272, 288)
167, 175 (cycle 336, 352)
199, 207 (cycle 400, 416)
231, 239 (cycle 464, 480)
These values don't make much sense, but it seems it should correlate to sprite pattern fetches

+PPU_Fetch_Counter.0{S10}
+PPU_Fetch_Counter.1{S13}
+PPU_Fetch_Counter.2{S18}
+PPU_Fetch_Counter.5{S25}
+PPU_Fetch_Counter.7{S29}

S4 NAND_5

R4

/Reset_Detect{R14}
Increment_Scanline_Fetch_Count{S6}

PPU_Fetch_Count_Match_0{R4}
+PPU_Cycle_336{R4}

Reg 2 DFF−Set

Q1 = 1 at PPU cycle 336 for incrementing V−split scanline counter.

Logic from S4:
Value 1x1xx111, when incrementing,
First triggered at 10100111=0xA7=167.
Because fetch count clocks the latch,
167+1 = 168. Since PPU has 2 cycles per fetch, 168*2=336 cycles.
This aligns with "unused NT fetches" at the end of the scanline.

+V−Split_Scanline_Counter.0{T9}
+V−Split_Scanline_Counter.1{T13}
+V−Split_Scanline_Counter.2{T16}
+V−Split_Scanline_Counter.3{T19}
+V−Split_Scanline_Counter.5{T25}
+V−Split_Scanline_Counter.6{T28}
+V−Split_Scanline_Counter.7{T31}

S30 NAND_7 /V−Split_Scanline=239{S30}

Scanline_End_Detected{T5}
T3 NAND_2 T2 INVERT T1 INVERT Reload_V−Split_Scanline_Count{T1}

0=prevent reload
1=allow reload

BH30 INVERT
BH29 GAL_21
BH28 INVERT
T4 D−FF−SR
BH31 NOR_3
BH32 INVERT

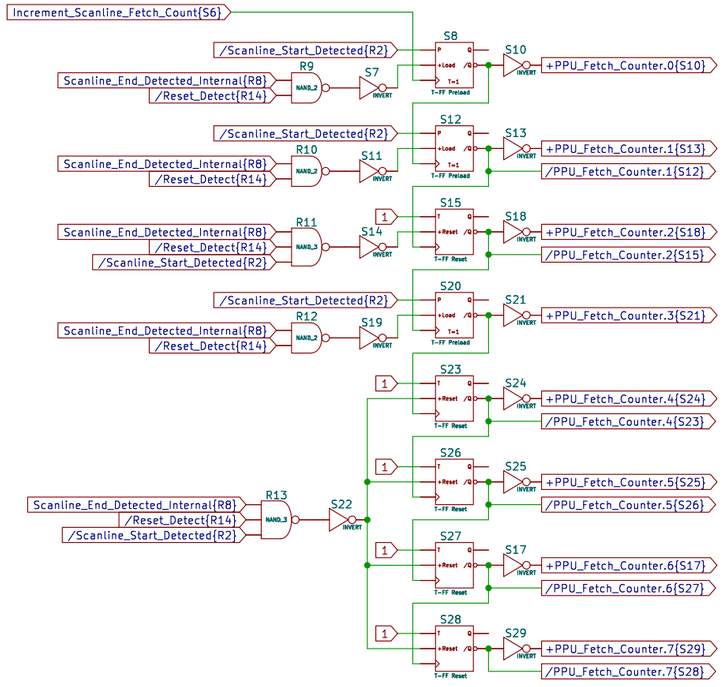
+Rendering_In_Frame{Z25}
/Reset_Detect{V1}
+Rendering_In_Frame{Z25}
Scanline_End_Detected{T5}

Sheet: /V−Split/
File: v_split.kicad_sch

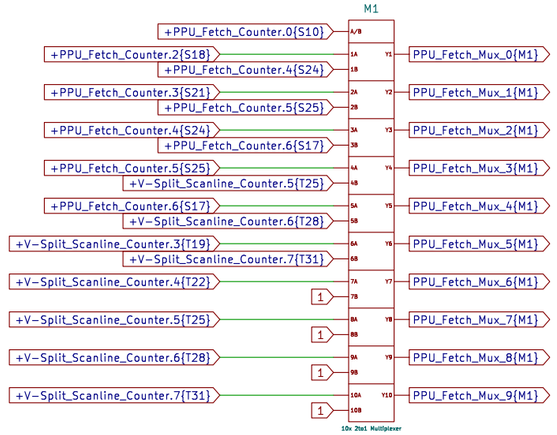
Title:

Size: B | Date: | **Rev:**
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PPU Fetch Counter

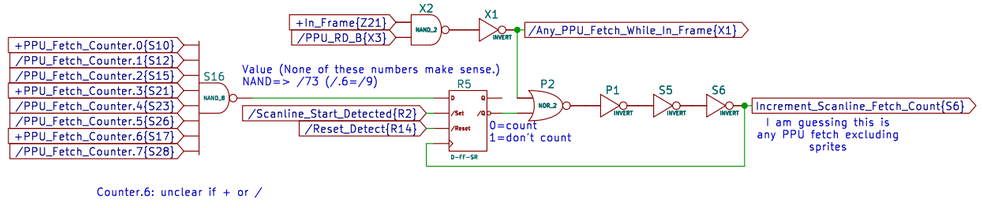


ExRAM Attribute Fetch Mux



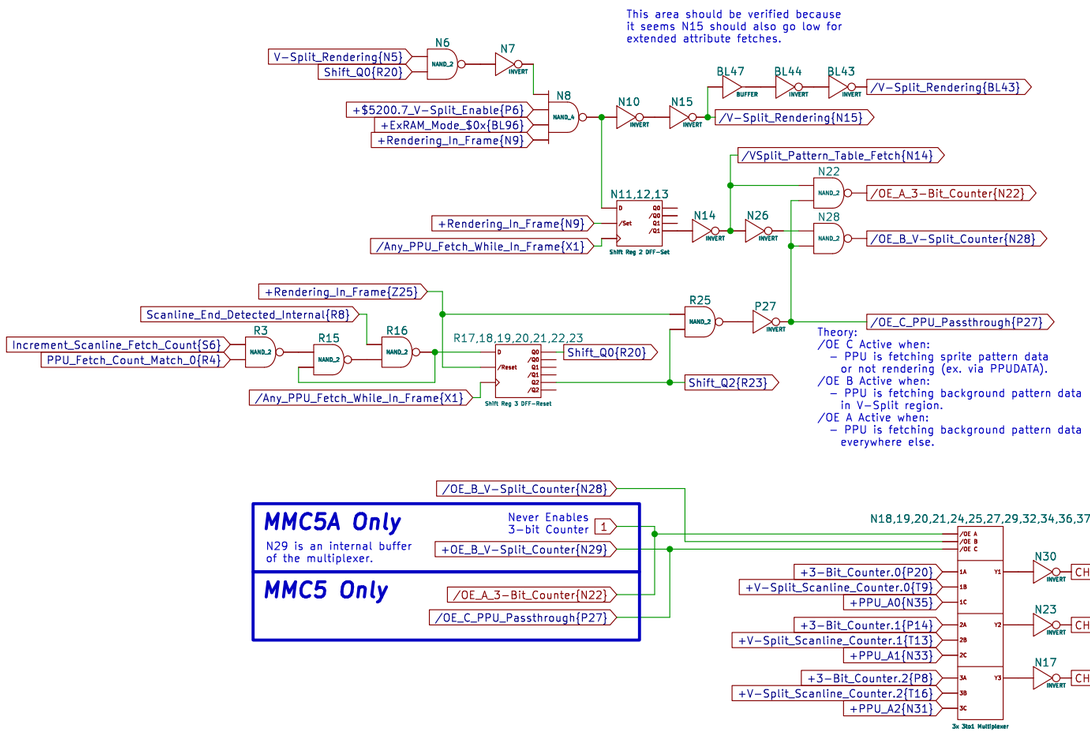
The output of this mux is the ExRAM address for the data to be substituted for the PPU.

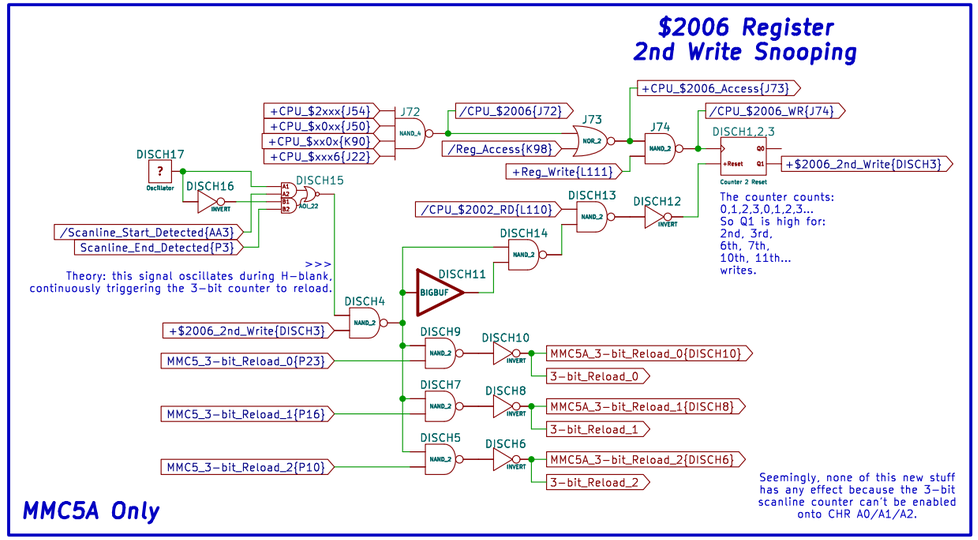
See sheet 14.

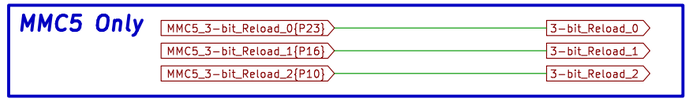


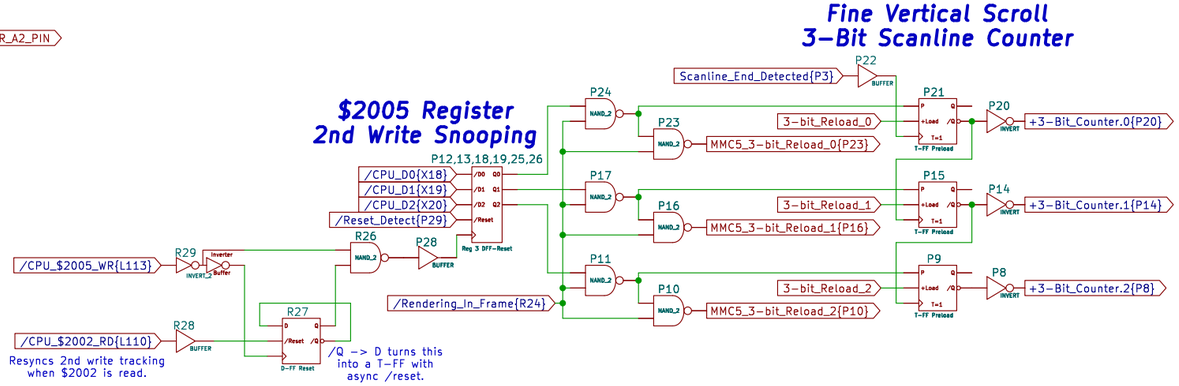
Counter.6: unclear if + or /

Sheet: /PPU Fetch Counting/	
File: ppu_fetch_counting.kicad_sch	
Title:	
Size: USLedger	Date:
KiCad E.D.A. 9.0.7	Id: 12/28









Sheet: /CHR A0,A1,A2/	
File: chr_a0_a1_a2.kicad_sch	
Title:	
Size: USLedger Date:	**Rev:**
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$5104 Register ExRAM Mode

CPU_D0_IN_PIN
CPU_D1_IN_PIN
/Reset_Detect{BL97}
/CPU_$5104_WR{L37}
BK82
Reg 2 DFF-Set
BK79
BK87
+ExRAM_Mode_%10{BK87}
BK80
BL49
/ExRAM_Mode_%01{BK80}
/ExRAM_Mode_%01{BL49}
BK81
BL96
/ExRAM_Mode_%00{BK81}
Appears to go nowhere.
+ExRAM_Mode_$0x{BL96}
Tracks whether $5104 has been written since reset detection
/Reset_Detect{RAX12}
M2_CLK_BUF1
+Invalid_A0_M2_Sequence{BN1}
Syncs SL3 to M2
BL98
D-FF Reset
BL89
SR Latch
BL90
BL91
BL94
BL92
BL93
BN2
BIGBUF
There appears to be a selectable delay jumper, connected at BM6.
BM10 BM9 BM8 BM7 BM6 BM5 BM4 BM3 BM2 BM1
BUFFER
BL88
BL85
BL84
BL83
PPU_RW_Mode_%0x_Access{BL83}
Not sure about polarity; this may be Mode %1x Access, and active low or high.
+ExRAM_Mode_$0x{BL96}
BL86
INVERT_3
+PPU_RD_A_PIN{PinInv}
+PPU_WR_PIN{PinInv}
BL82
BL87
BL81
BIGBUF

PPU is fetching from nametable or attribute ($2000+) and V-Split is not rendering and ExRAM mode is %0x (a PPU mode)
+PPU_A13_A{BK97}
+ExRAM_Mode_$0x{BL96}
BK94
BK95
+Drive_ExRAM_Ax_From_PPU_Ax{BK95}
/V-Split_Rendering{N15}
BK92
PPU is fetching from nametable or attribute ($2000+) and V-Split is rendering and ExRAM mode is %0x (a PPU mode)
BK93
BK91
+Drive_ExRAM_Ax_For_Substituted_Fetch{BK91}
PPU is fetching from pattern table (below $2000) or ExRAM mode is %1x (a CPU mode)
BK96
BJ51
+Drive_ExRAM_Ax_From_CPU_Ax{BJ51}
Note:
Maybe N15 means /Extended Attribute Fetch OR V-Split Rendering?
N15 seems to directly depend on V-Split enable register bit though.

BJ50 ExRAM_A0{BJ50} PPU_Fetch_Mux_0{M1} PPU_A0_Pin
BJ49 ExRAM_A1{BJ49} PPU_Fetch_Mux_1{M1} PPU_A1_Pin
BJ48 ExRAM_A2{BJ48} PPU_Fetch_Mux_2{M1} PPU_A2_Pin
BJ47 ExRAM_A3{BJ47} PPU_Fetch_Mux_3{M1} PPU_A3_Pin
BJ46 ExRAM_A4{BJ46} PPU_Fetch_Mux_4{M1} PPU_A4_Pin
BJ45 ExRAM_A5{BJ45} PPU_Fetch_Mux_5{M1} PPU_A5_Pin
BJ44 ExRAM_A6{BJ44} PPU_Fetch_Mux_6{M1} PPU_A6_Pin
BJ43 ExRAM_A7{BJ43} PPU_Fetch_Mux_7{M1} PPU_A7_Pin
BJ42 ExRAM_A8{BJ42} PPU_Fetch_Mux_8{M1} PPU_A8_Pin
BJ41 ExRAM_A9{BJ41} PPU_Fetch_Mux_9{M1} PPU_A9_Pin
NAND Mux 3

BJ40
CPU_A0_Pin CPU_A1_Pin CPU_A2_Pin CPU_A3_Pin CPU_A4_Pin CPU_A5_Pin CPU_A6_Pin CPU_A7_Pin CPU_A8_Pin CPU_A9_Pin
Reg 10 D-Latch

+CPU_ExRAM_Access{L117}
+Reg_Write{BK88}
BK85
/CPU_Write_To_ExRAM_Any_Mode{BK85}
BK86
+ExRAM_Mode_%10{BK87}
Writing to ExRAM via $5C00-5FFF in mode %10
Writing to an ExRAM nametable via PPU via $2000
+ExRAM_Nametable_Access{A9}
+ExRAM_Mode_$0x{BL96}
+PPU_WR_PIN{PinInv}
BK83
BK84
+ExRAM_WR{BK84}
This is the write signal for the ExRAM itself.

BN4
BIGBUF
BL105
BL106
BL109
+ExRAM_Mode_$0x{BL96}
/CPU_Write_To_ExRAM_Any_Mode{BK85}
/Q{BL109}
BL108
BL99
/Q{BL99}
+PPU_A13_A{BK97}
BL102
/Reset_Detect{RAX12}
BL101
BL100
BN3
BIGBUF
BL107
+PPU_RD_A_PIN{PinInv}

/Q{BL109}
/CPU_Write_To_ExRAM_Any_Mode{BK85}
+ExRAM_Mode_$0x{BL96}
BK89
DAL_21
MMC5 Only
+PPU_RD_A_PIN{PinInv}
MMC5A Only
/Q{BL99}
BK90
+ExRAM_Mode_$0x{BL96}

CPU_D0_IN_PIN CPU_D1_IN_PIN CPU_D2_IN_PIN CPU_D3_IN_PIN CPU_D4_IN_PIN CPU_D5_IN_PIN CPU_D6_IN_PIN CPU_D7_IN_PIN
BK78
Reg 8 D-Latch
BK77
CPU_D0_In_Latched{BK77} CPU_D1_In_Latched{BK77} CPU_D2_In_Latched{BK77} CPU_D3_In_Latched{BK77} CPU_D4_In_Latched{BK77} CPU_D5_In_Latched{BK77} CPU_D6_In_Latched{BK77} CPU_D7_In_Latched{BK77}
Reg 8 D-Latch

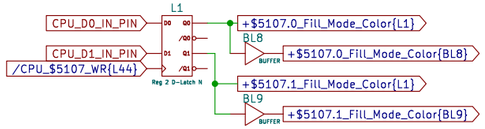
ExRAM_D0_Read ExRAM_D1_Read ExRAM_D2_Read ExRAM_D3_Read ExRAM_D4_Read ExRAM_D5_Read ExRAM_D6_Read ExRAM_D7_Read
BJ12(a) BJ12(b) BJ12(c) BJ12(d) BJ12(e) BJ12(f) BJ12(g) BJ12(h)
BUFFER
ExRAM_D0_Read{BJ12(a)} ExRAM_D1_Read{BJ12(b)} ExRAM_D2_Read{BJ12(c)} ExRAM_D3_Read{BJ12(d)} ExRAM_D4_Read{BJ12(e)} ExRAM_D5_Read{BJ12(f)} ExRAM_D6_Read{BJ12(g)} ExRAM_D7_Read{BJ12(h)}
BJ11
/ExRAM_Read_D0_Latch{BJ11} /ExRAM_Read_D1_Latch{BJ11} /ExRAM_Read_D2_Latch{BJ11} /ExRAM_Read_D3_Latch{BJ11} /ExRAM_Read_D4_Latch{BJ11} /ExRAM_Read_D5_Latch{BJ11} +ExRAM_Read_D6_Latch{BJ11} +ExRAM_Read_D7_Latch{BJ11}
+PPU_RD_A{BJ8}
+PPU_Fetch_Count.1{S13}
+PPU_Fetch_Count.0{S10}
BJ9
BJ10
BJ7
Reg 8 D-Latch N

/Scanline_Counter_Reset{Z25}
+ExRAM_Mode_$0x{BL96}
BL95
BL103
BL104
BJ39b
Inverter
INVERT_3
Buffer

CPU_D0_In_Latched{BK77} PPU_D0_In{BigBuf}
CPU_D1_In_Latched{BK77} PPU_D1_In{BigBuf}
CPU_D2_In_Latched{BK77} PPU_D2_In{BigBuf}
CPU_D3_In_Latched{BK77} PPU_D3_In{BigBuf}
CPU_D4_In_Latched{BK77} PPU_D4_In{BigBuf}
CPU_D5_In_Latched{BK77} PPU_D5_In{BigBuf}
CPU_D6_In_Latched{BK77} PPU_D6_In{BigBuf}
CPU_D7_In_Latched{BK77} PPU_D7_In{BigBuf}
BJ39 ExRAM_D0_Write{BJ39}
BJ38 ExRAM_D1_Write{BJ38}
BJ37 ExRAM_D2_Write{BJ37}
BJ36 ExRAM_D3_Write{BJ36}
BJ35 ExRAM_D4_Write{BJ35}
BJ34 ExRAM_D5_Write{BJ34}
BJ33 ExRAM_D6_Write{BJ33}
BJ32 ExRAM_D7_Write{BJ32}
NAND Mux 2

Sheet: /ExRAM Mode/
File: exram_mode.kicad_sch
Title:
Size: USLedger | Date:
KiCad E.D.A. 9.0.7
Rev:
Id: 14/28

$5107 Register
Fill-Mode Color



$5106 Register
Fill-Mode Tile



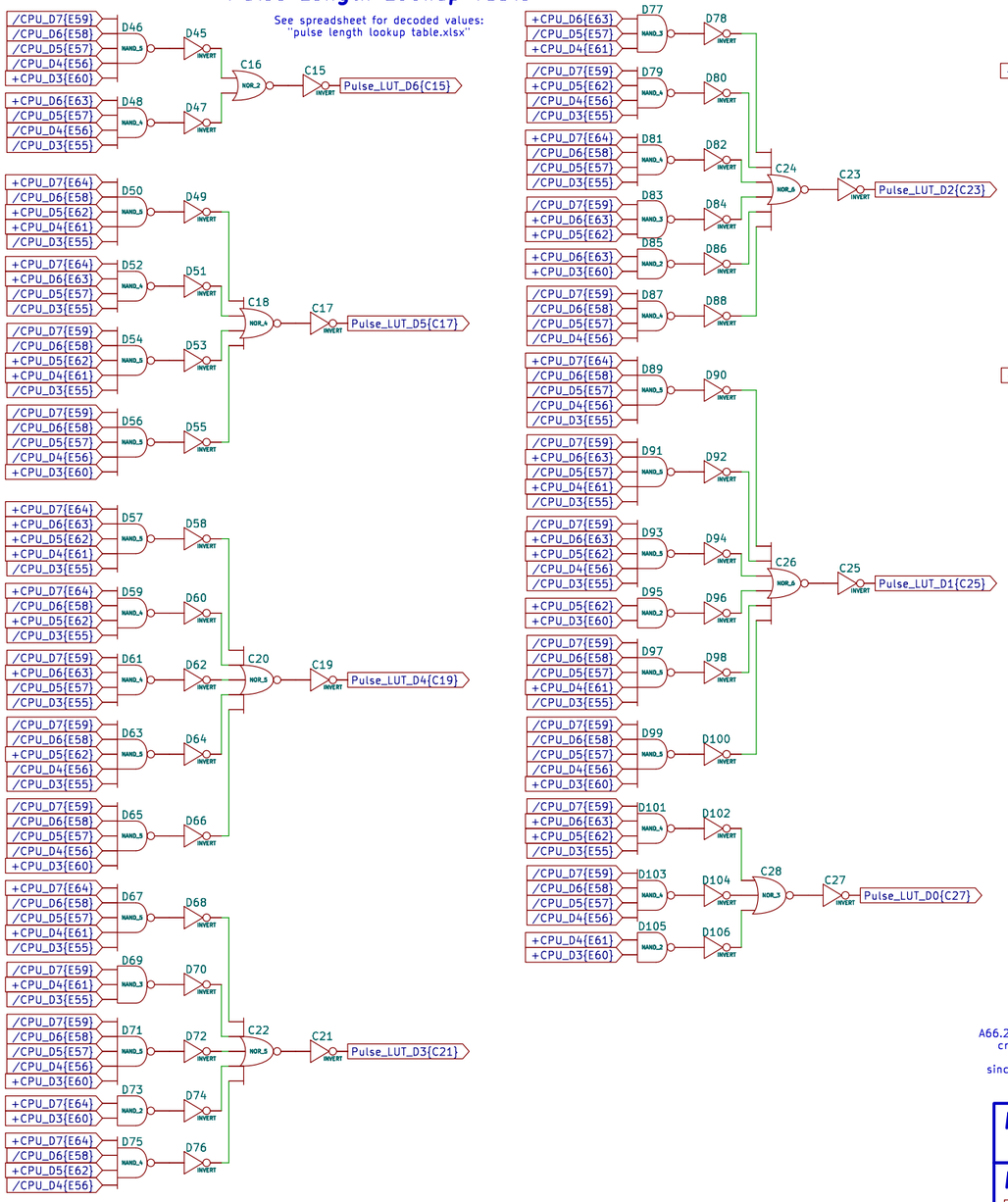
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File: fill_mode.kicad_sch

Title:

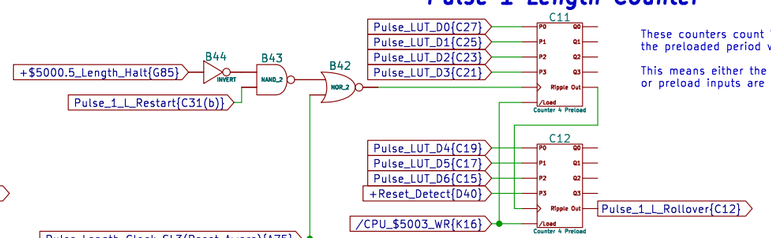
Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 15/28

5-Bit to 7-Bit Pulse Length Lookup Table

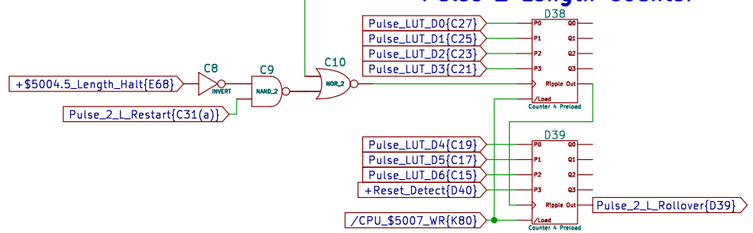
See spreadsheet for decoded values:
"pulse length lookup table.xlsx"



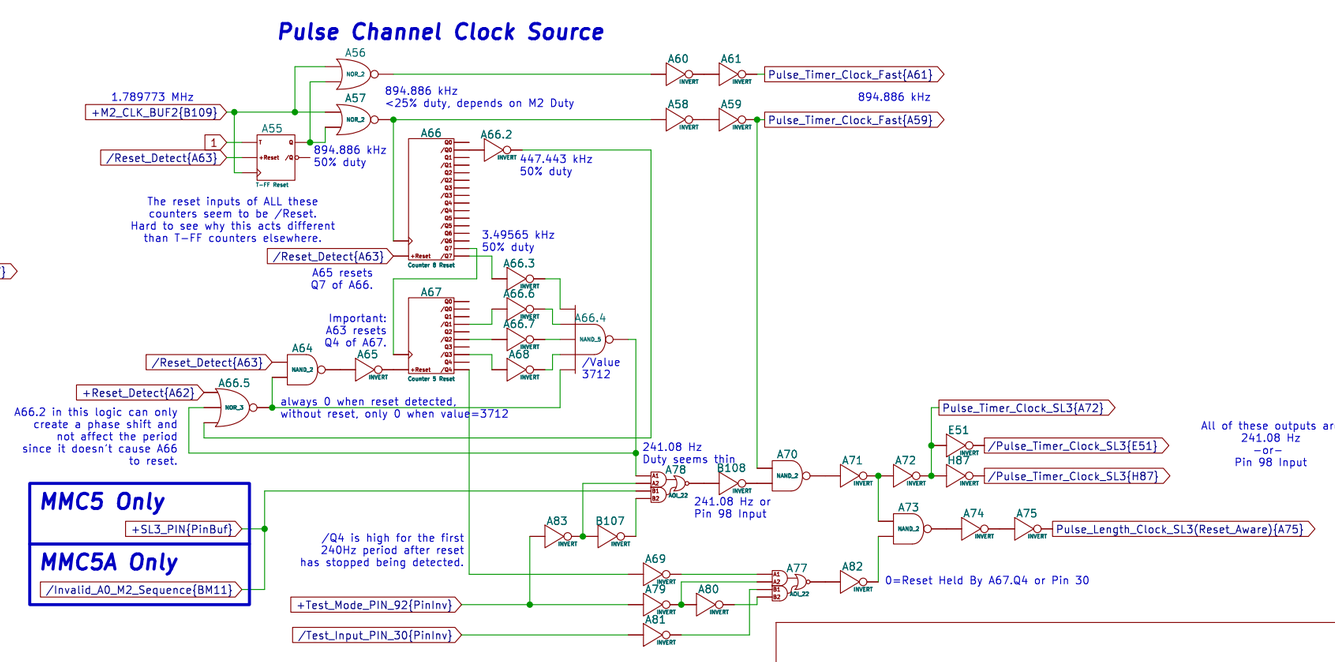
$5003 Register
Pulse 1 Length Counter



$5007 Register
Pulse 2 Length Counter



Pulse Channel Clock Source

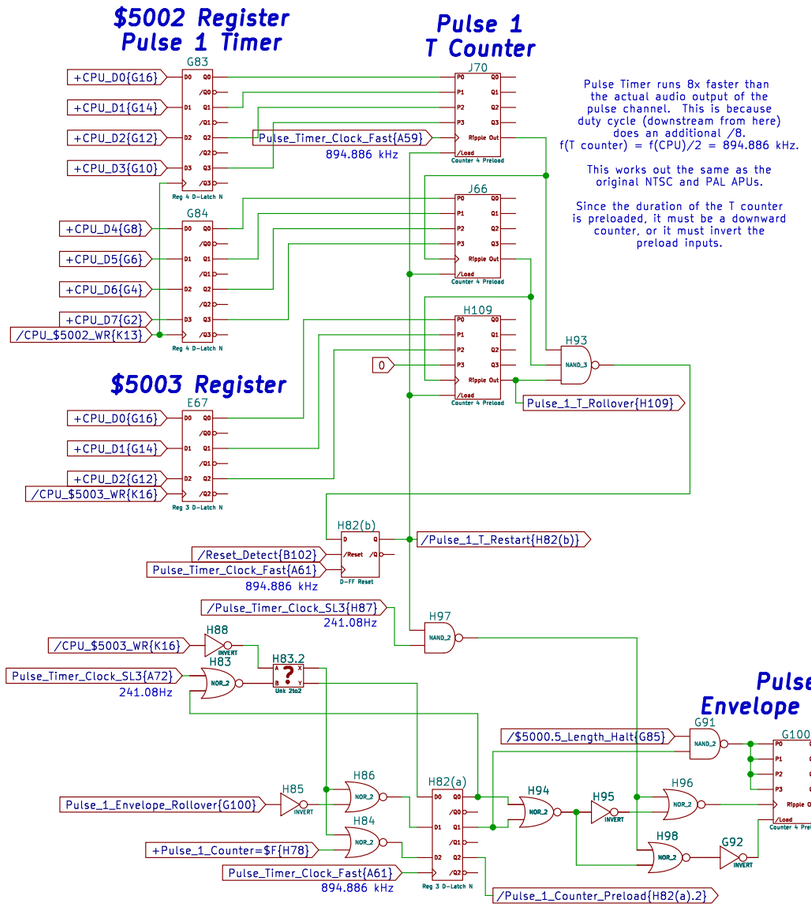


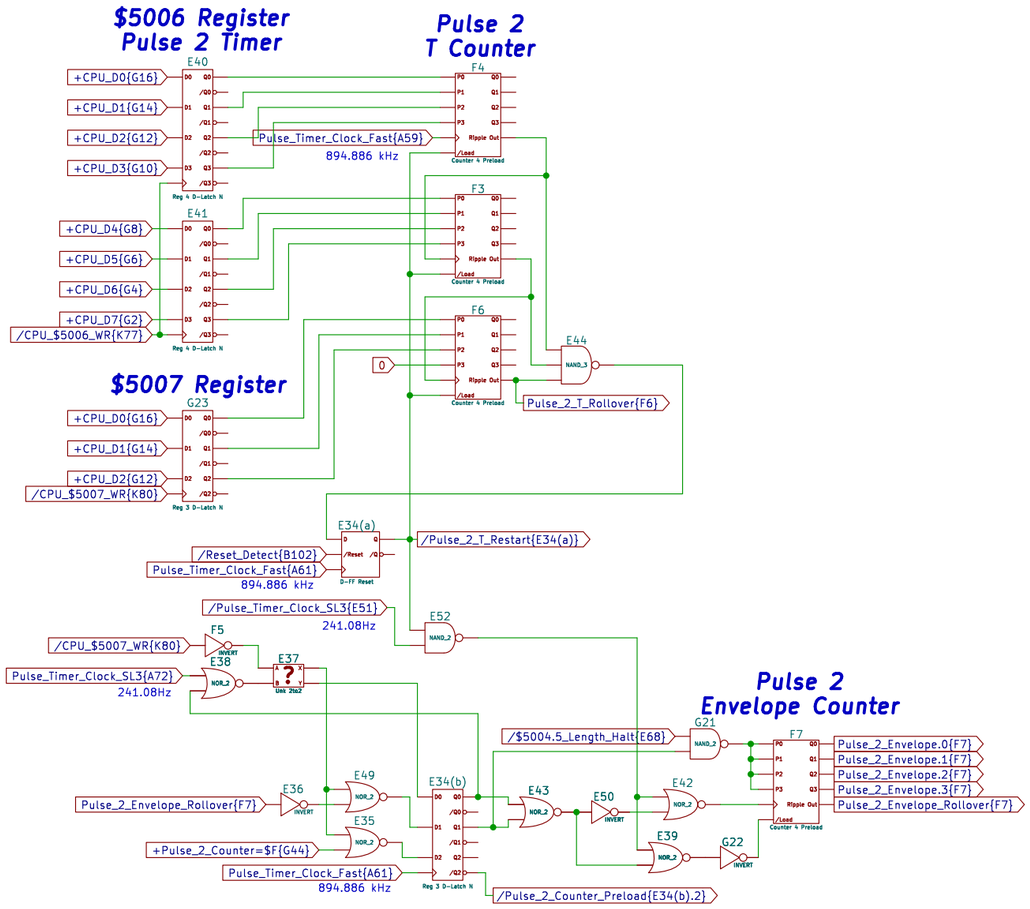
Sheet: /Pulse Length Lookup Table/
File: pulse_length_lookup_table.kicad_sch

Title:

Size: USLedger | Date: | **Rev:**

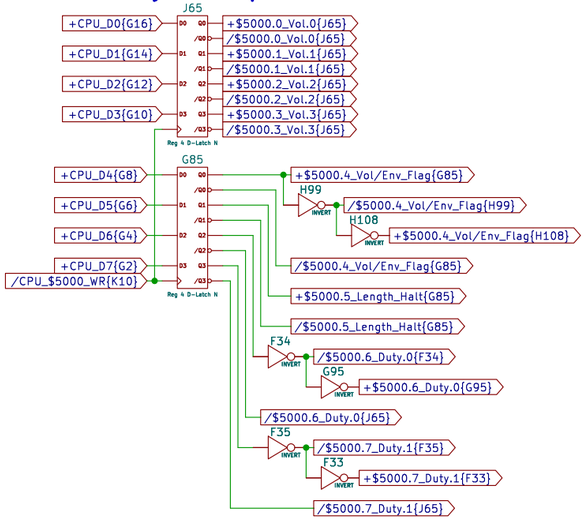
KiCad E.D.A. 9.0.7 | Id: 16/28



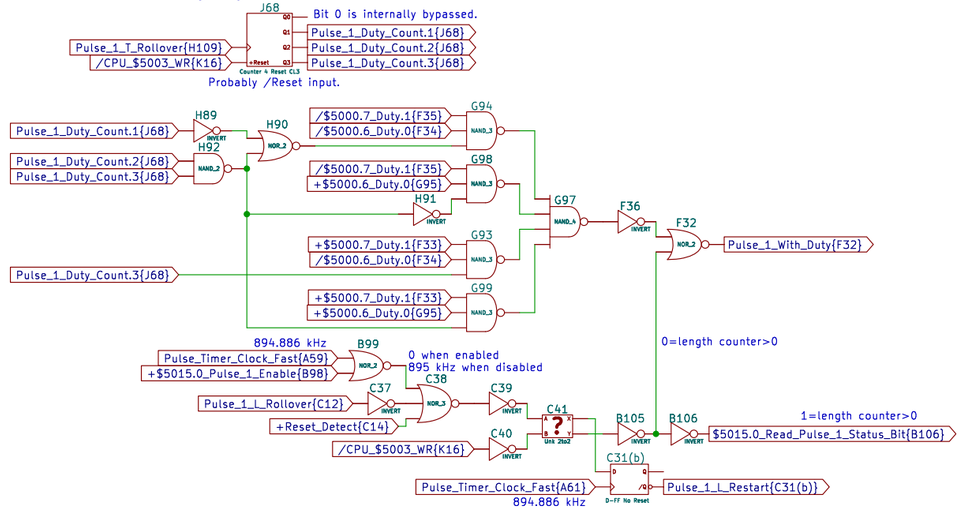


Sheet: /Pulse Timer/	
File: pulse_timer.kicad_sch	
Title:	
Size: USLedger | Date:	**Rev:**
KiCad E.D.A. 9.0.7	Id: 17/28

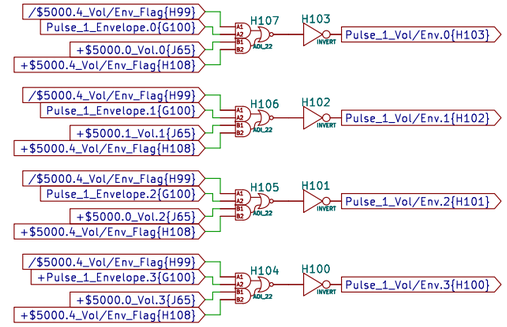
$5000 Register
Pulse 1 Duty Envelope Volume



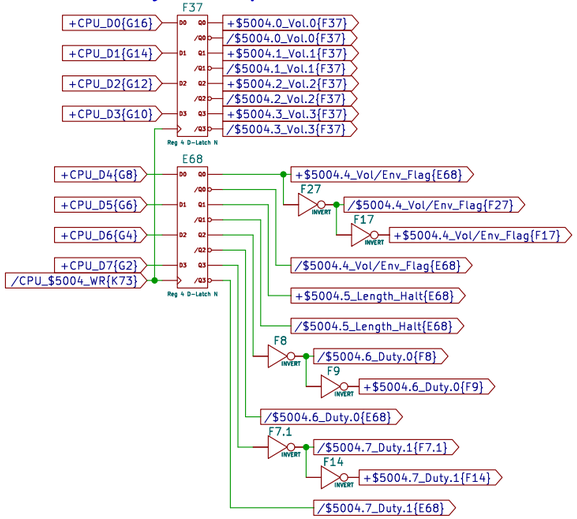
Pulse 1
Duty Cycle Counter



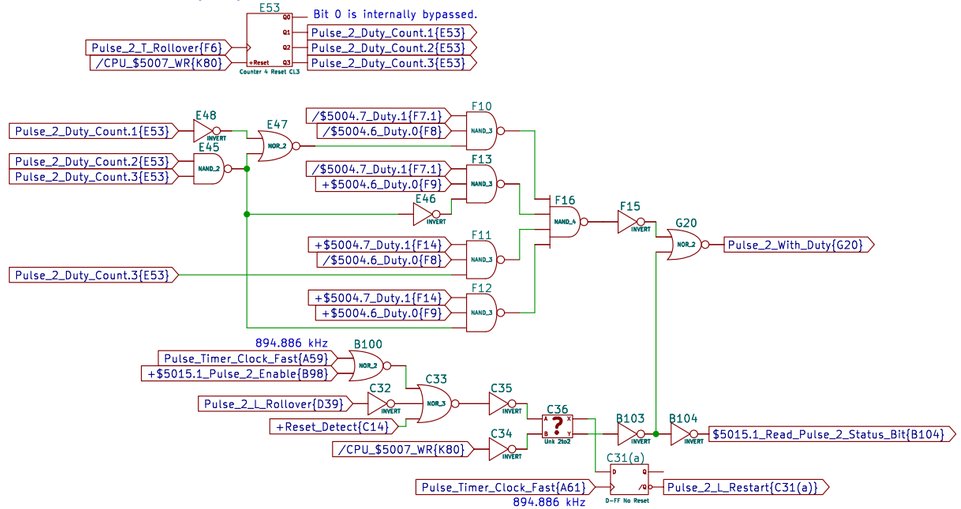
Pulse 1
Envelope/Volume Multiplexing



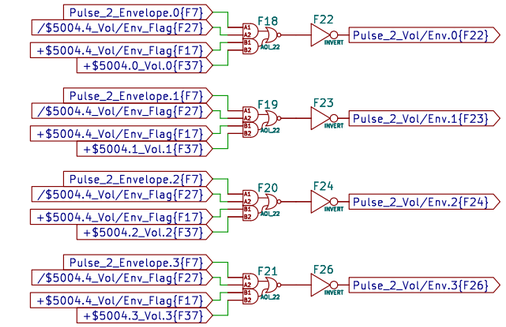
$5004 Register
Pulse 2 Duty Envelope Volume



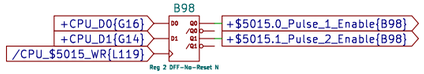
Pulse 2
Duty Cycle Counter



Pulse 2
Envelope/Volume Multiplexing



$5015 Write Register
Pulse 1&2 Status

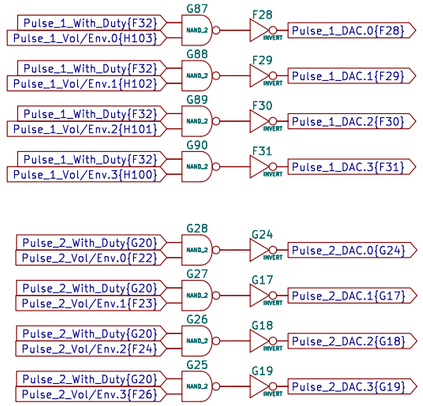


Sheet: /Pulse Duty Envelope Volume/
File: pulse_duty_envelope_volume.kicad_sch

Title:

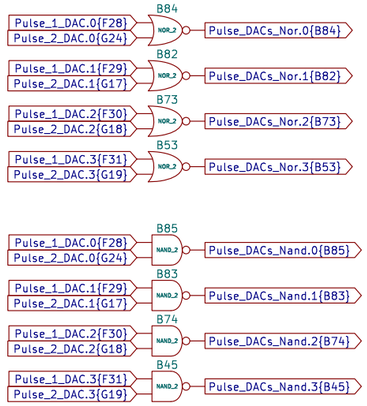
Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 18/28

Pulse 1&2
Individual 4-bit DAC Outputs

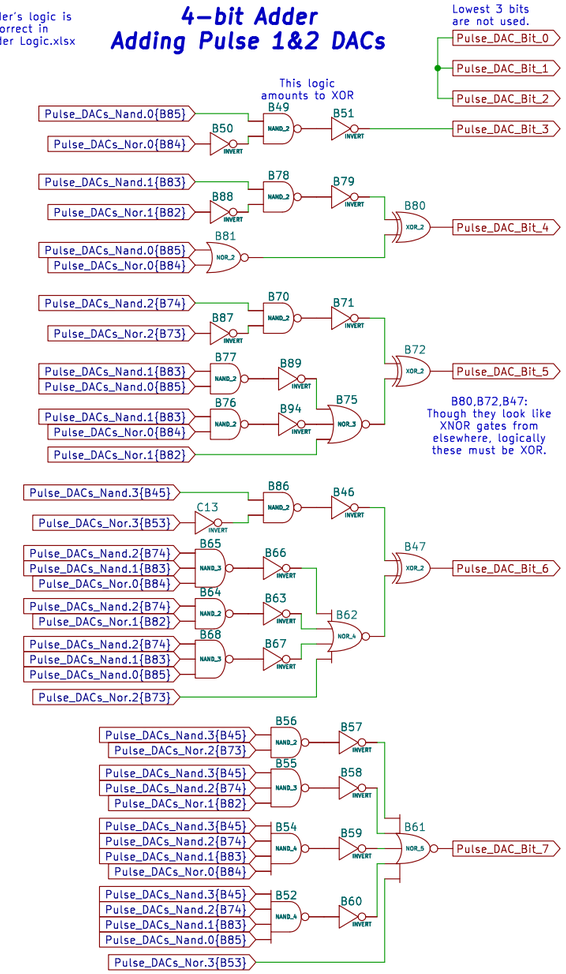


Components for
4-bit Adder

Note: The adder's logic is verified correct in Pulse DAC Adder Logic.xlsx



4-bit Adder
Adding Pulse 1&2 DACs



Sheet: /Pulse DAC/
File: Pulse DAC.kicad_sch

Title:

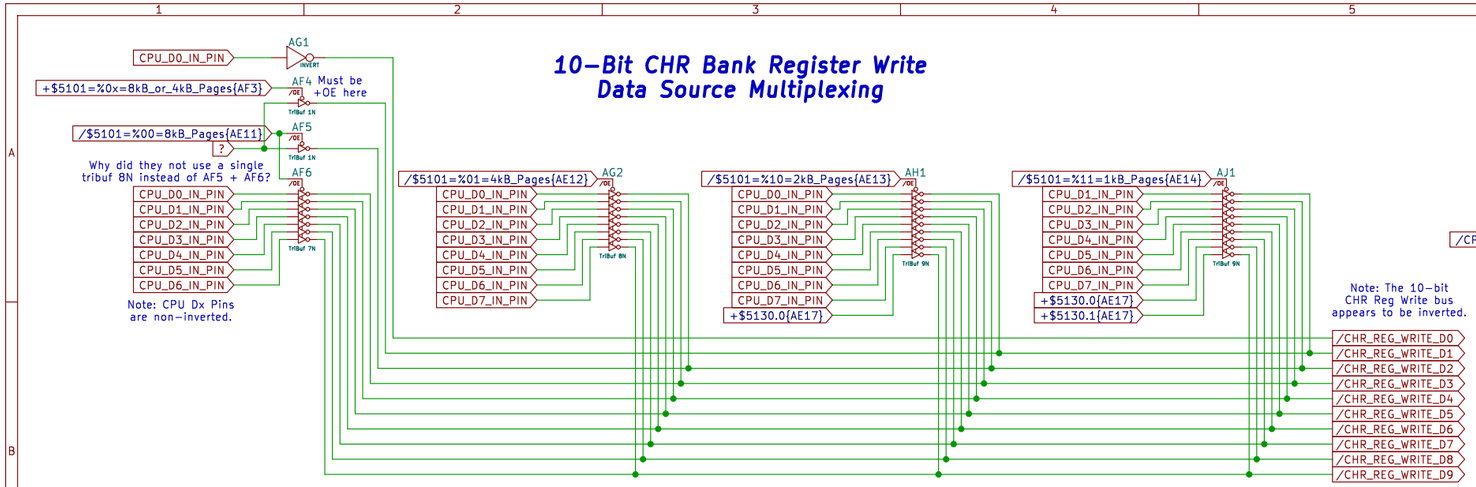
Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 19/28

PPU_D0_IN_PIN
PPU_D1_IN_PIN
PPU_D2_IN_PIN
PPU_D3_IN_PIN
PPU_D4_IN_PIN
PPU_D5_IN_PIN
PPU_D6_IN_PIN
PPU_D7_IN_PIN
{BigBuf}
BIGBUF
PPU_D0_In{BigBuf}
PPU_D1_In{BigBuf}
PPU_D2_In{BigBuf}
PPU_D3_In{BigBuf}
PPU_D4_In{BigBuf}
PPU_D5_In{BigBuf}
PPU_D6_In{BigBuf}
PPU_D7_In{BigBuf}
/V-Split_Rendering{BL43}
/Scanline_Counter_Reset{Z25}
/ExRAM_Mode_%01{BL49}
/Extended Attributes Enabled
+Fill_Mode_Nametable_Access{BL33}
/PPU_Attribute_Access{BL25}
+ExRAM_Nametable_Access{A9}
+ExRAM_Mode_$0x{BL48}
BL10
BL34
BL35
BL40
BL32
BL28
BL36
BL37
BL38
BL29
BL30
BL39
BL31
BL41
BL42
BL46
BL45
BK39
BK40
BK42
BK44
BK41
BK43
+$5107.1_Fill_Mode_Color{BL9}
+$5107.0_Fill_Mode_Color{BL8}
BH25
BH27
BH26
+PPU_Fetch_Counter.3{S21}
+V-Split_Scanline_Counter.4{T22}
BH22
BH24
BH20
BH18
BH21
BH23
BH19
BH17
BH4
BH5
BH6
BH7
BH8
BH3
BH2
BH1
ExRAM_D7_Read{BJ12(h)}
ExRAM_D5_Read{BJ12(f)}
ExRAM_D3_Read{BJ12(d)}
ExRAM_D1_Read{BJ12(b)}
+ExRAM_Read_D7_Latch{BJ11}
+ExRAM_Read_D6_Latch{BJ11}
BH12
BH13
BH14
BH15
BH16
BH11
BH10
BH9
ExRAM_D6_Read{BJ12(g)}
ExRAM_D4_Read{BJ12(e)}
ExRAM_D2_Read{BJ12(c)}
ExRAM_D0_Read{BJ12(a)}
ExRAM_D0_Read{BJ12(a)}
+$5106.0_Fill_Mode_Tile{L11}
ExRAM_D1_Read{BJ12(b)}
+$5106.1_Fill_Mode_Tile{L12}
ExRAM_D2_Read{BJ12(c)}
+$5106.2_Fill_Mode_Tile{L14}
ExRAM_D3_Read{BJ12(d)}
+$5106.3_Fill_Mode_Tile{L13}
ExRAM_D4_Read{BJ12(e)}
+$5106.4_Fill_Mode_Tile{L18}
ExRAM_D5_Read{BJ12(f)}
+$5106.5_Fill_Mode_Tile{L17}
ExRAM_D6_Read{BJ12(g)}
+$5106.6_Fill_Mode_Tile{L16}
ExRAM_D7_Read{BJ12(h)}
+$5106.7_Fill_Mode_Tile{L15}
BK1
BL12
BL1
BL13
BL2
BL14
BL3
BL15
BL4
BL16
BL5
BL17
BL6
BL18
BL7
BL19
BK7
BK11
BK15
BK19
BK23
BK27
BK31
BK35
BK10
BK8
BK14
BK12
BK18
BK16
BK22
BK20
BK26
BK24
BK30
BK28
BK34
BK32
BK38
BK36
BK9
BK13
BK17
BK21
BK25
BK29
BK33
BK37
PPU_D0_OUT_PIN
PPU_D1_OUT_PIN
PPU_D2_OUT_PIN
PPU_D3_OUT_PIN
PPU_D4_OUT_PIN
PPU_D5_OUT_PIN
PPU_D6_OUT_PIN
PPU_D7_OUT_PIN
CIRAM_CE_PIN{A9}
+PPU_RD_A{BJ1}
+PPU_A13_B{BK6}
/PPU_A12_B{BK3}
BK2
BK5
BK4
PPU_Dx_Direction{BK4}

Sheet: /PPU Data Bus/
File: ppu_data_bus.kicad_sch

Title:

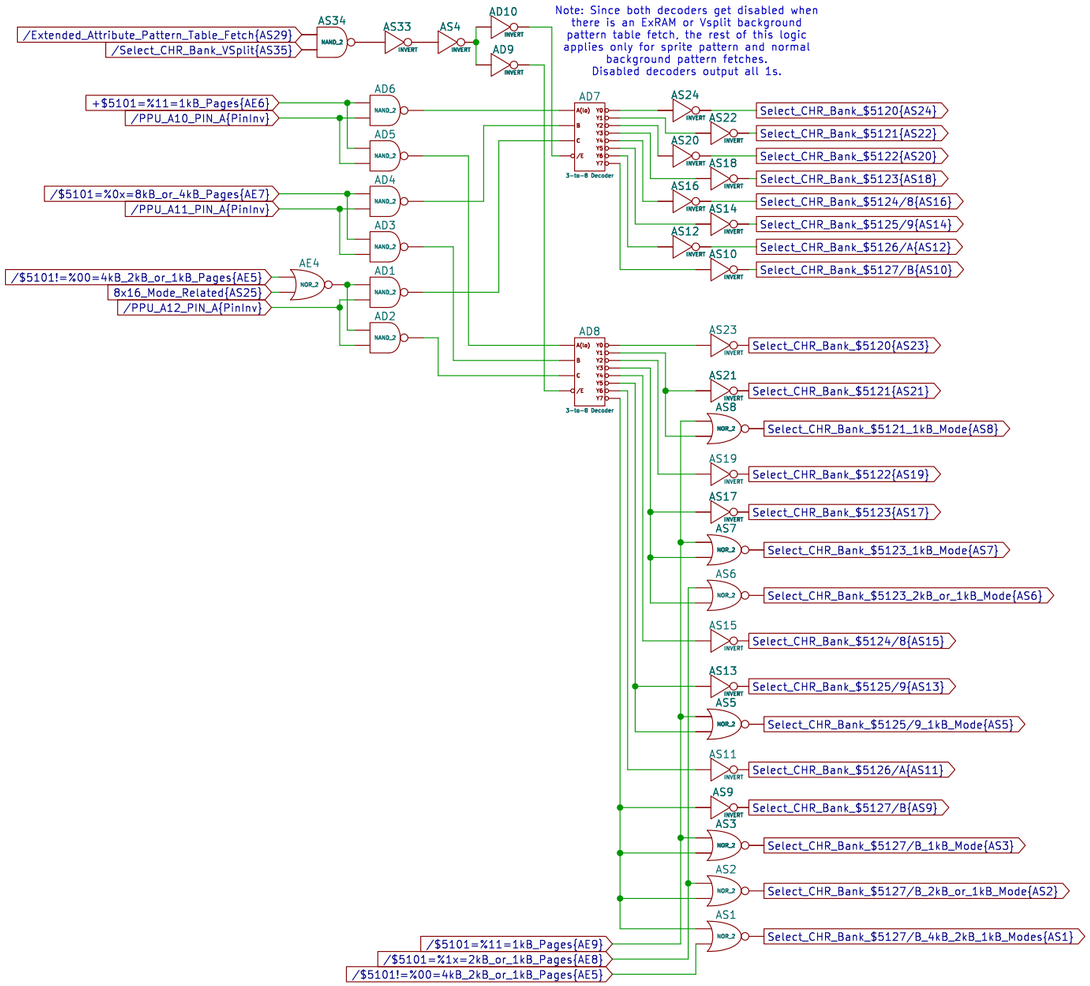
Size: USLedger | Date: | **Rev:**
KiCad E.D.A. 9.0.7 | Id: 20/28



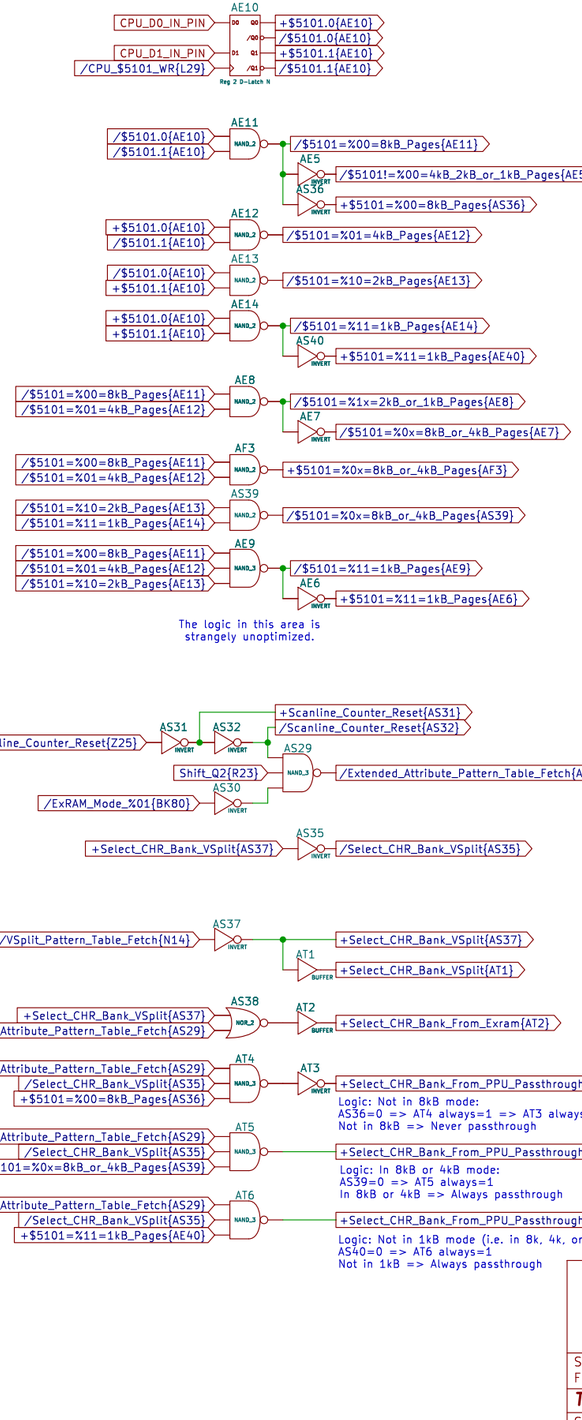
Sheet: /CHR Bank Registers/
File: chr_bank_registers.kicad_sch

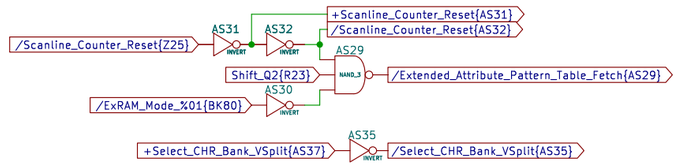
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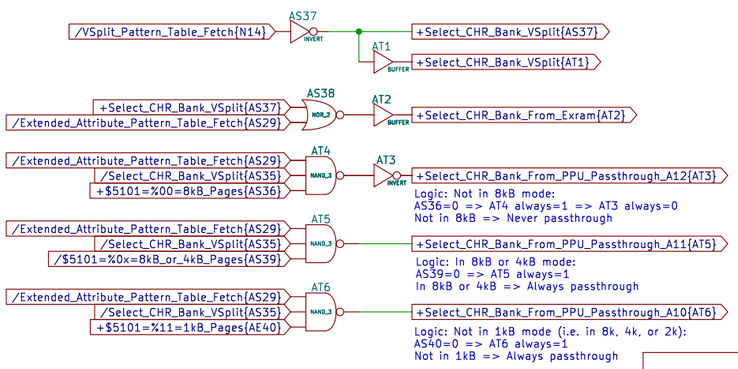
Size: USLedger | Date: | **Rev:**
KiCad E.D.A. 9.0.7 | Id: 21/28



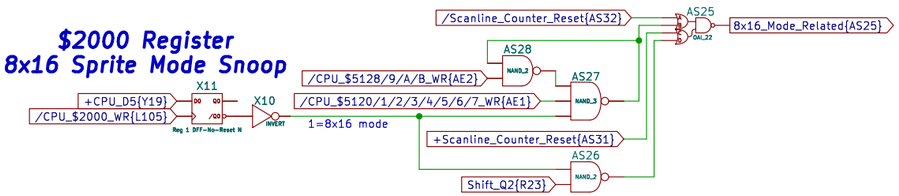
$5101 Register
CHR Mode







$2000 Register
8x16 Sprite Mode Snoop

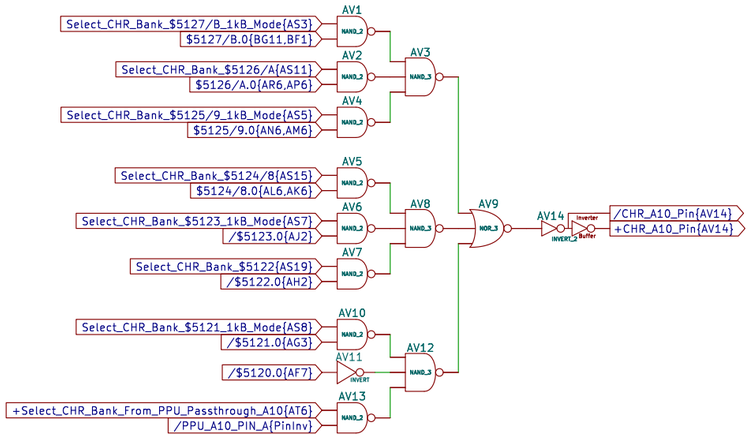


Sheet: /CHR Bank Selection Logic/
File: chr_bank_selection_logic.kicad_sch

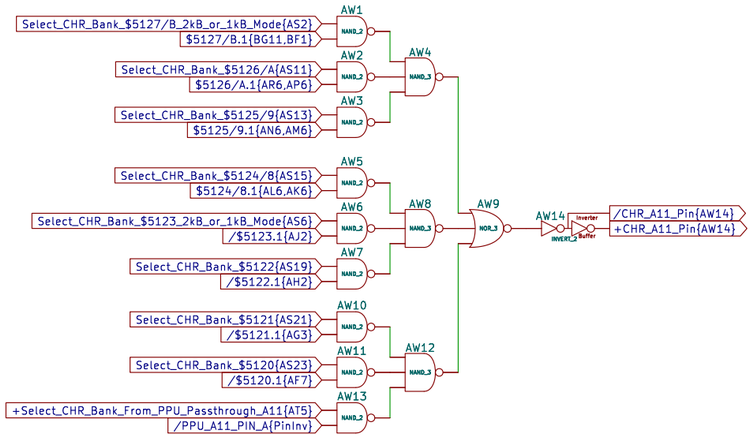
Title:

Size: USLedger | Date: | **Rev:**
KiCad E.D.A. 9.0.7 | Id: 22/28

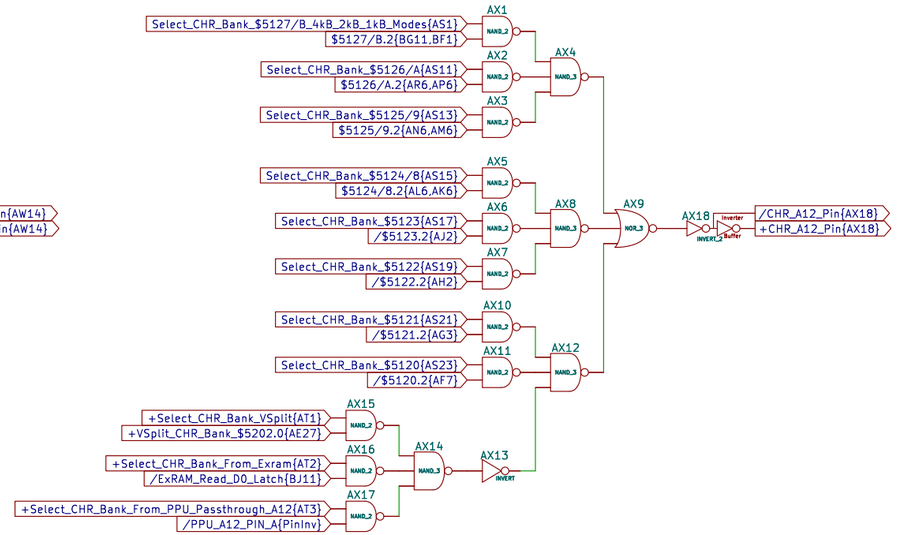
CHR A10



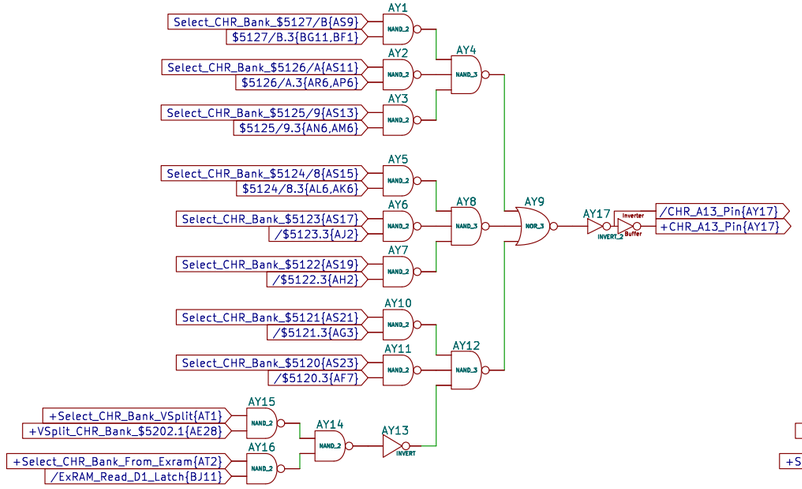
CHR A11



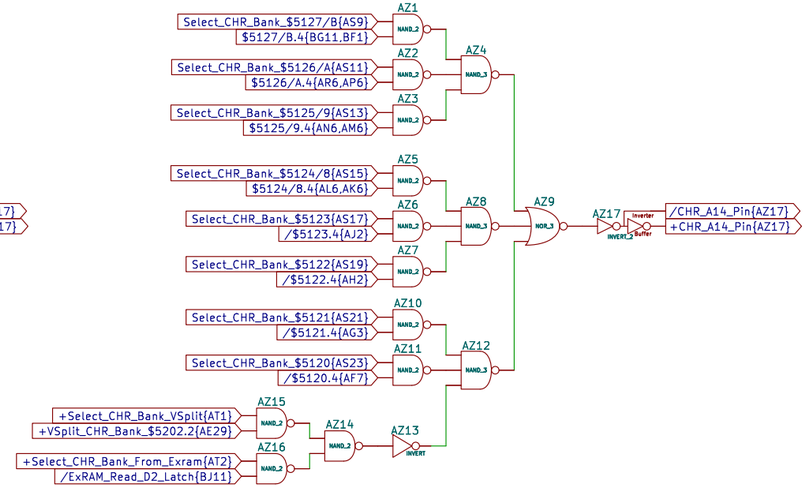
CHR A12



CHR A13



CHR A14

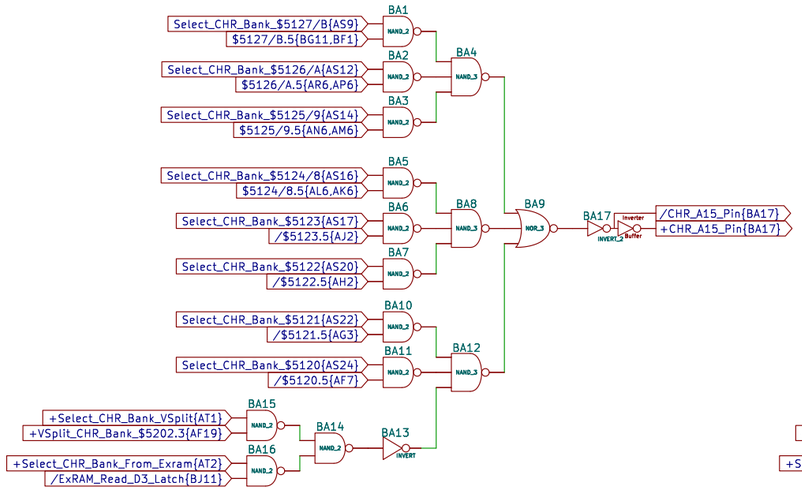


Sheet: /CHR Address Bus 10 to 14/
File: chr_address_bus_10_to_14.kicad_sch

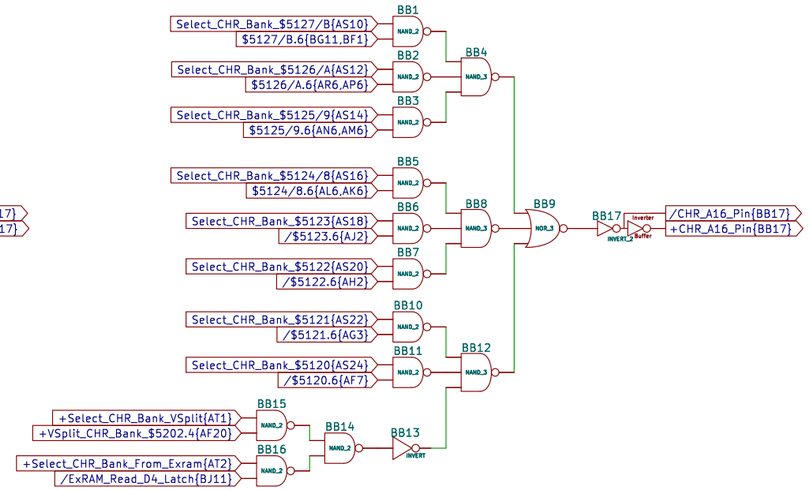
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Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 23/28

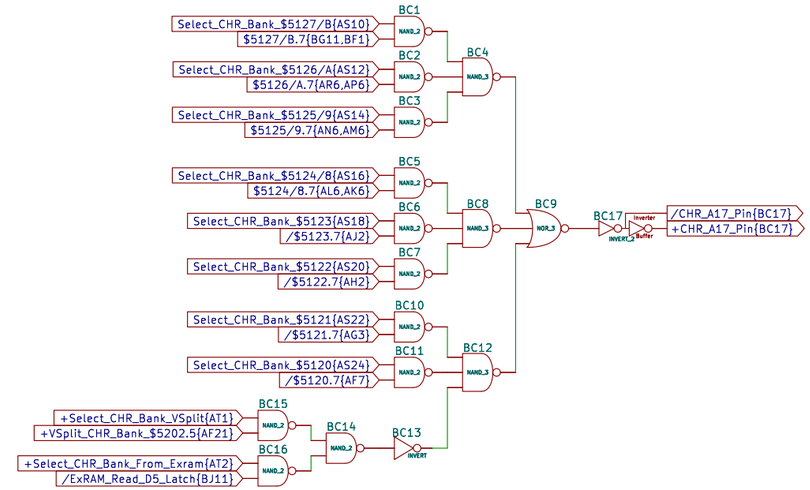
CHR A15



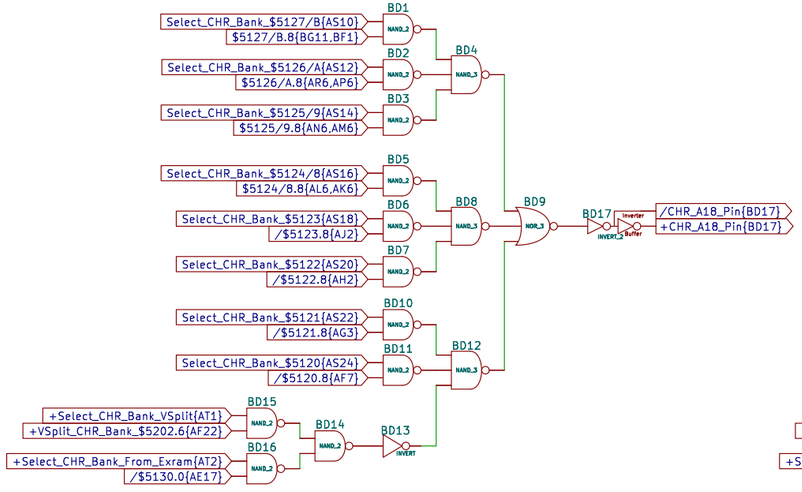
CHR A16



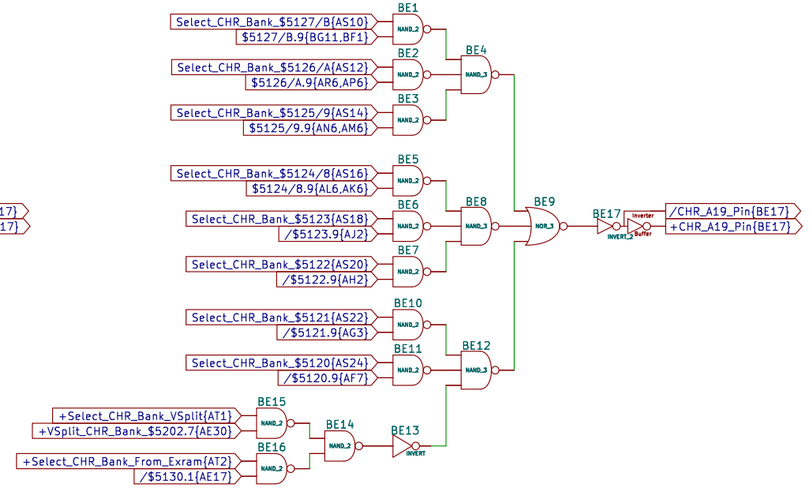
CHR A17



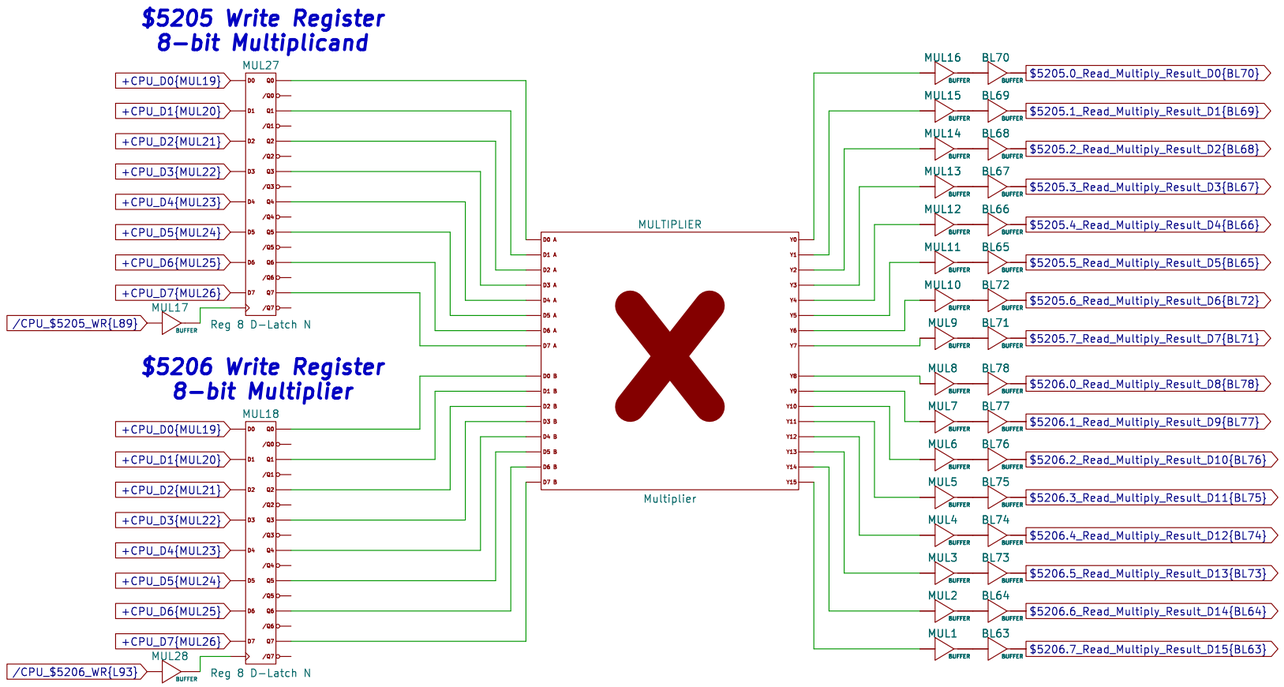
CHR A18



CHR A19



Sheet: /CHR Address Bus 15 to 19/		
File: chr_address_bus_15_to_19.kicad_sch		
Title:		
Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 24/28

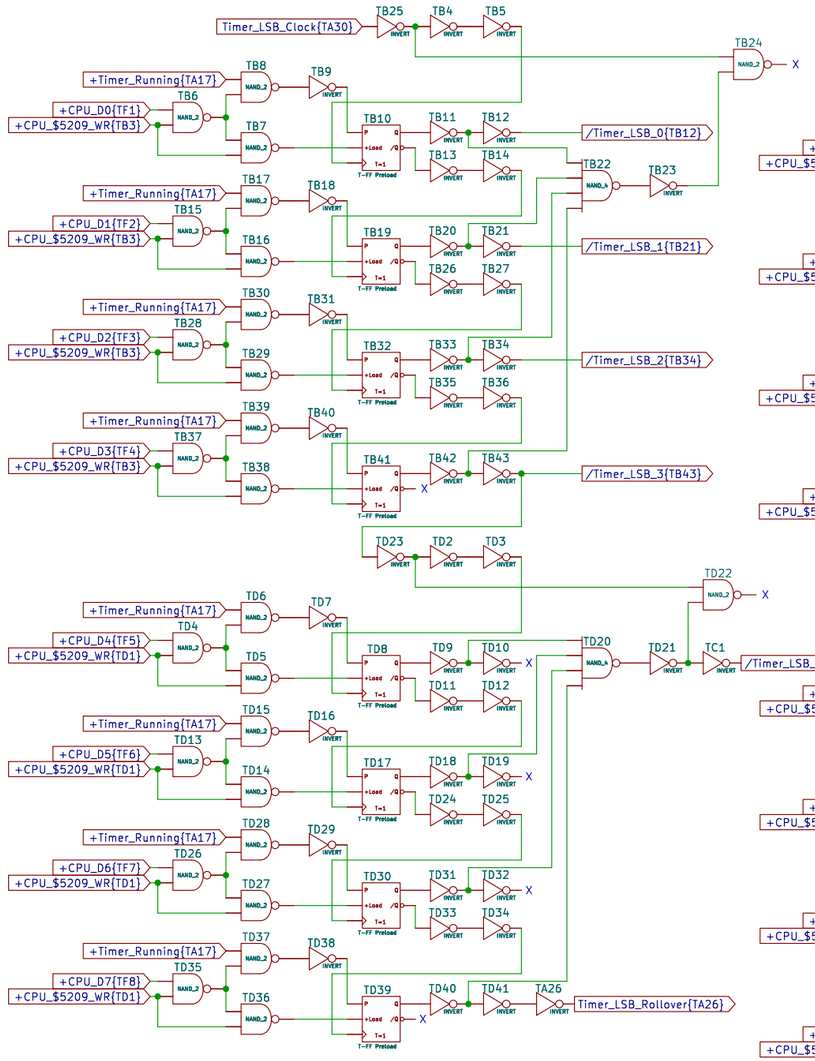


Sheet: /Multiplier/
File: multiplier.kicad_sch

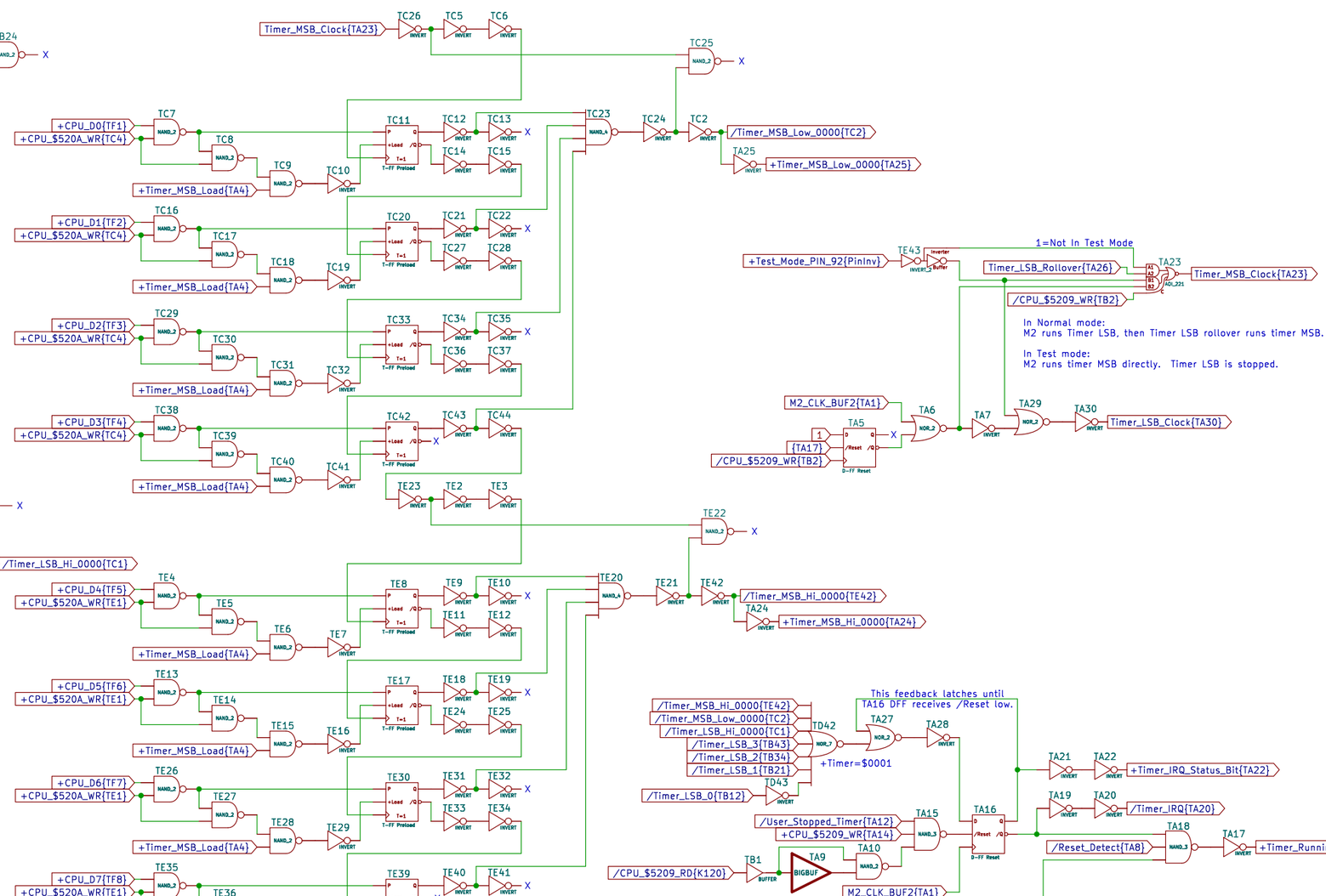
Title:

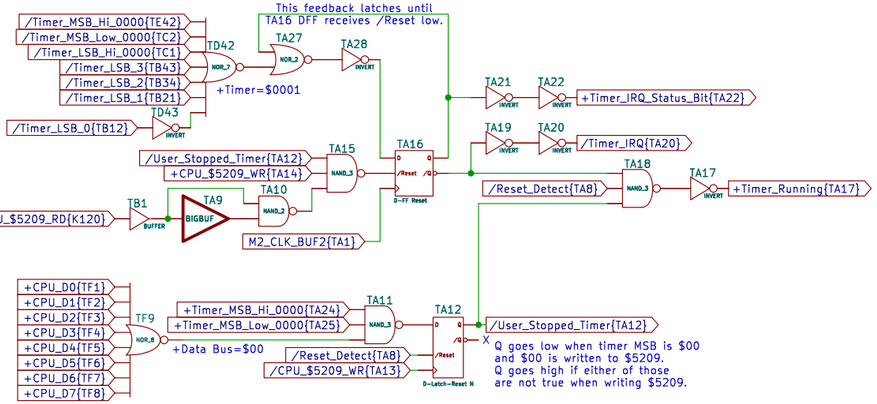
Size: USLedger | Date: | **Rev:**
KiCad E.D.A. 9.0.7 | Id: 25/28

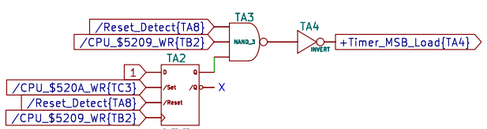
Timer LSB

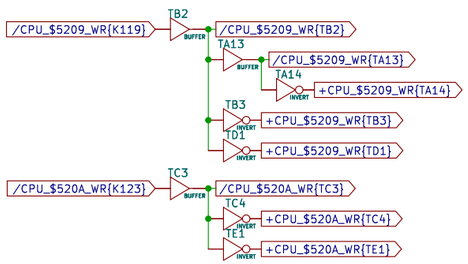


Timer MSB

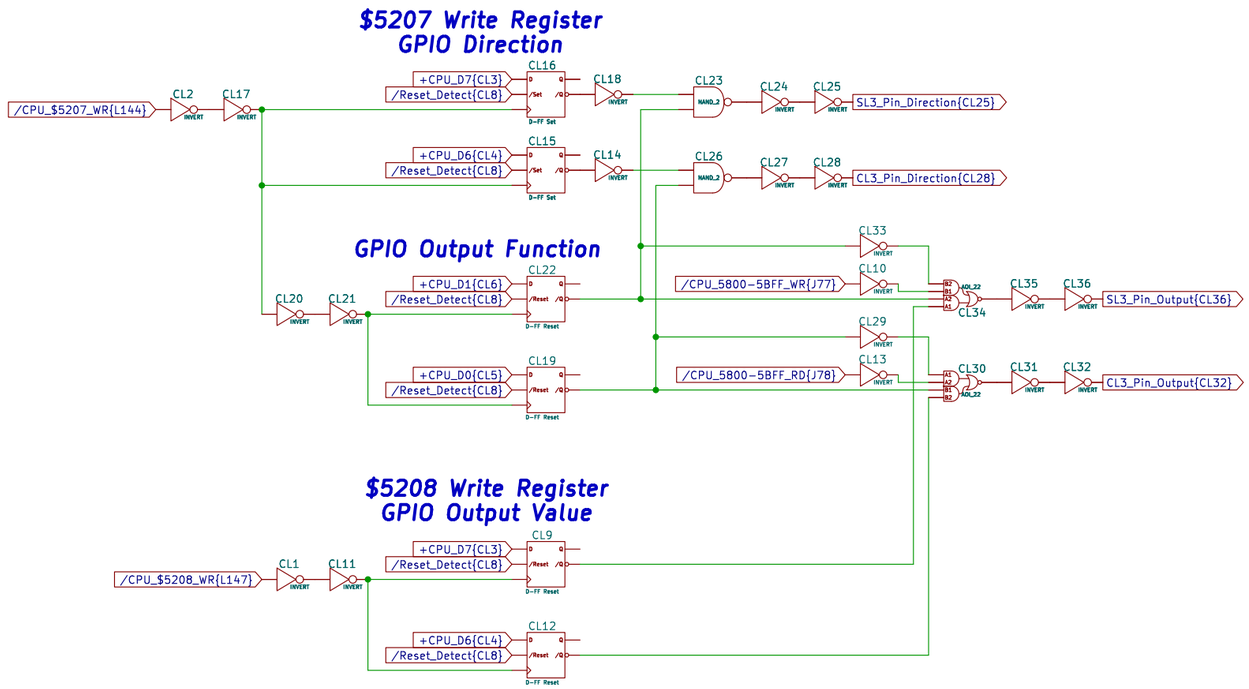








Sheet: /MMC5A Timer/		
File: mmc5a_timer.kicad_sch		
Title:		
Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 26/28

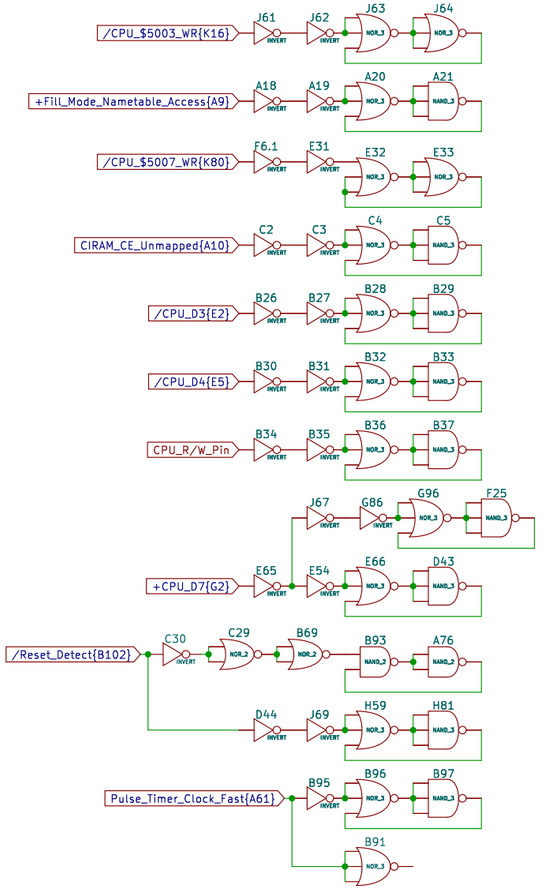


Sheet: /MMC5A CL3 SL3 GPIO/
File: mmc5a_cl3_sl3_gpio.kicad_sch

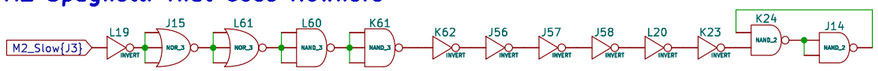
Title:

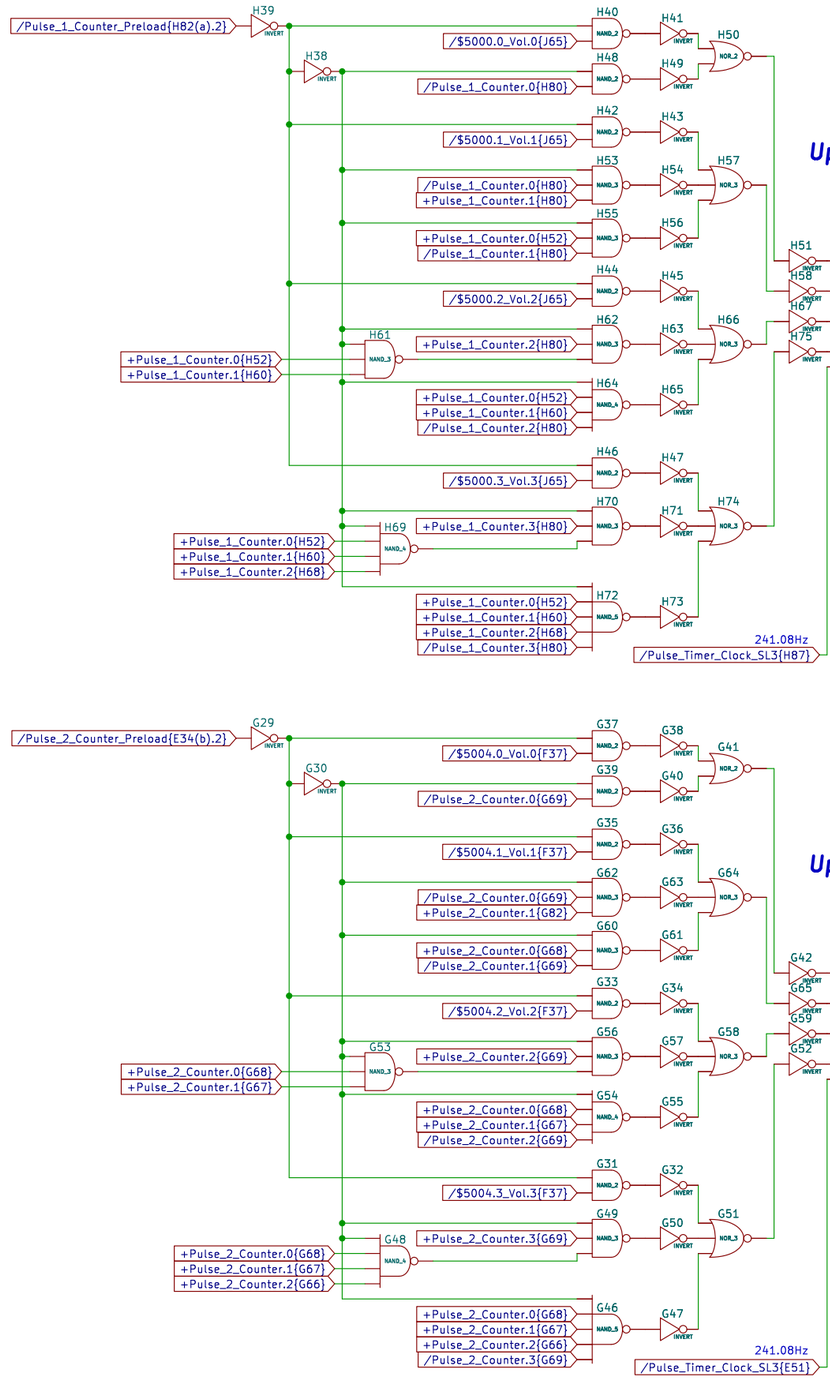
Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 27/28

This sheet contains all residual items that don't serve any apparent purpose.



M2 Spaghetti That Goes Nowhere





Pulse 1 Unused 4-bit Binary Up-Counter With Preload

The only output that goes elsewhere is H78, which gets latched by H82(a), coming directly back here as /Pulse_1_Counter_Preload; all self-contained. It does not appear to provide any other external signals, so I don't think it actually does anything. It seems like an abandoned envelope counter that was left in.

It latches the 4-bit pulse volume, then counts up at a rate of 240 Hz. Nothing stops it from counting or rolling over from $F back to $0.

Pulse 2 Unused 4-bit Binary Up-Counter With Preload

Sheet: /Artifacts/
File: artifacts.kicad_sch

Title:

Size: USLedger | Date: | **Rev:**
KiCad E.D.A. 9.0.7 | Id: 28/28